

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DskEth.sil etc.

Creation date: 23-Feb-79 3:28:26

For: Spruce

48 total sheets = 47 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Character code 57b not found in font GATES32.

Character w/ code 117b, font Helvetica7BI would lie off the page.

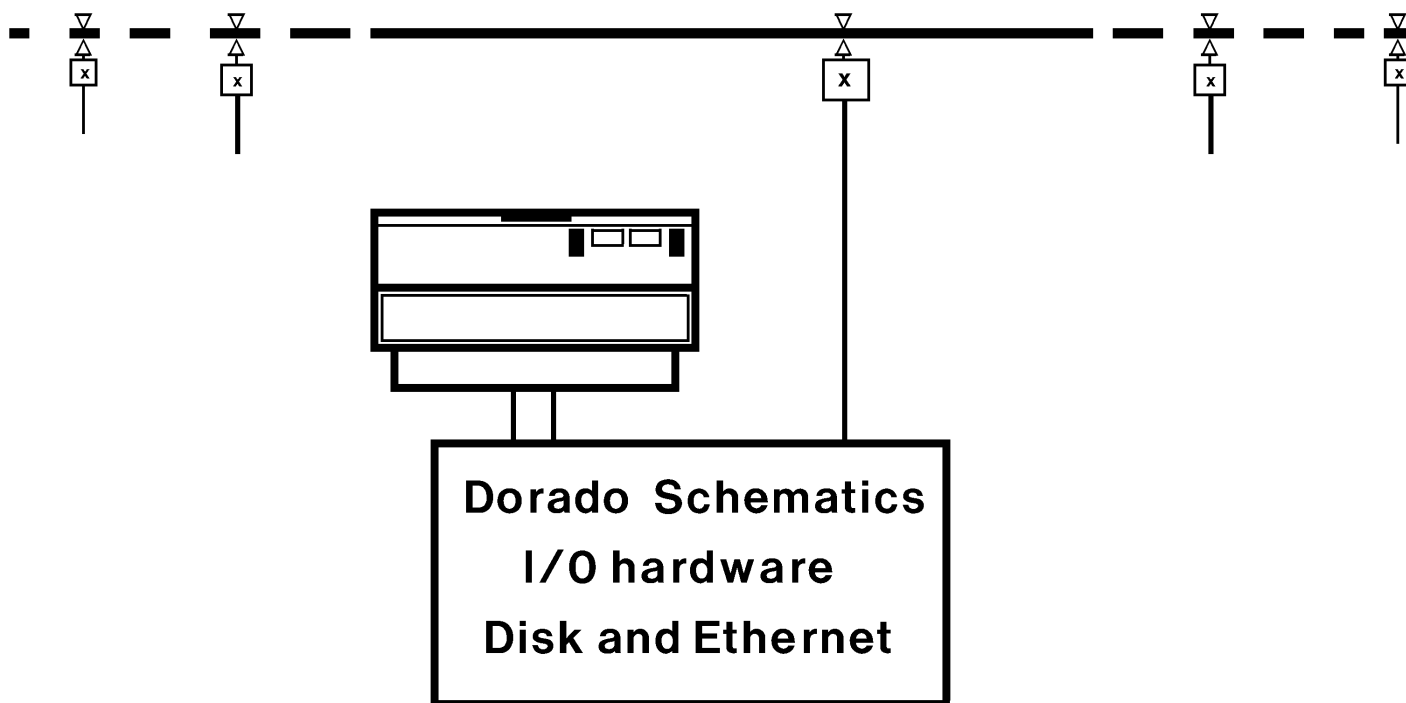
Character w/ code 117b, font Helvetica7BI would lie off the page.

Character w/ code 117b, font Helvetica7BI would lie off the page.

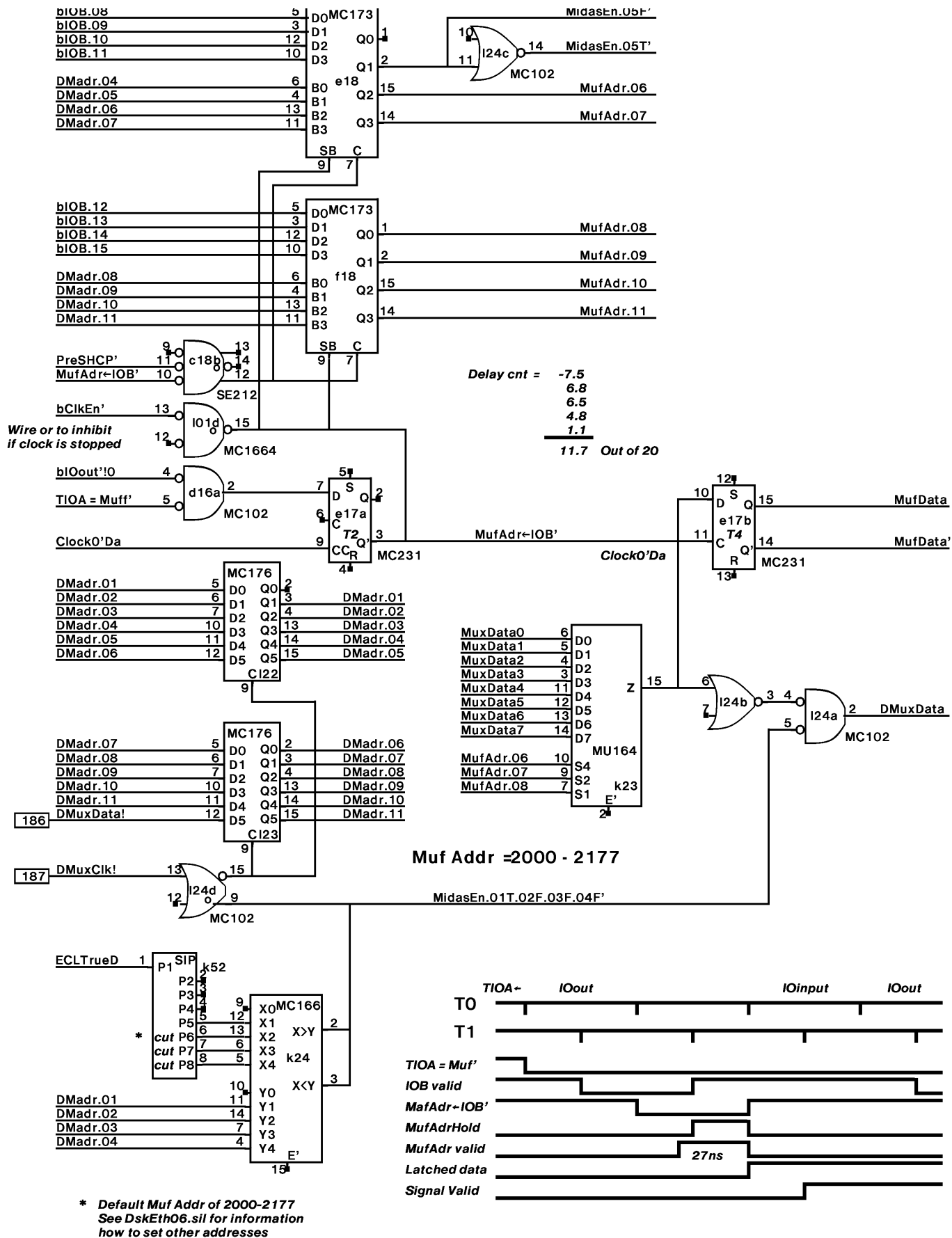
Character w/ code 124b, font Helvetica7BI would lie off the page.

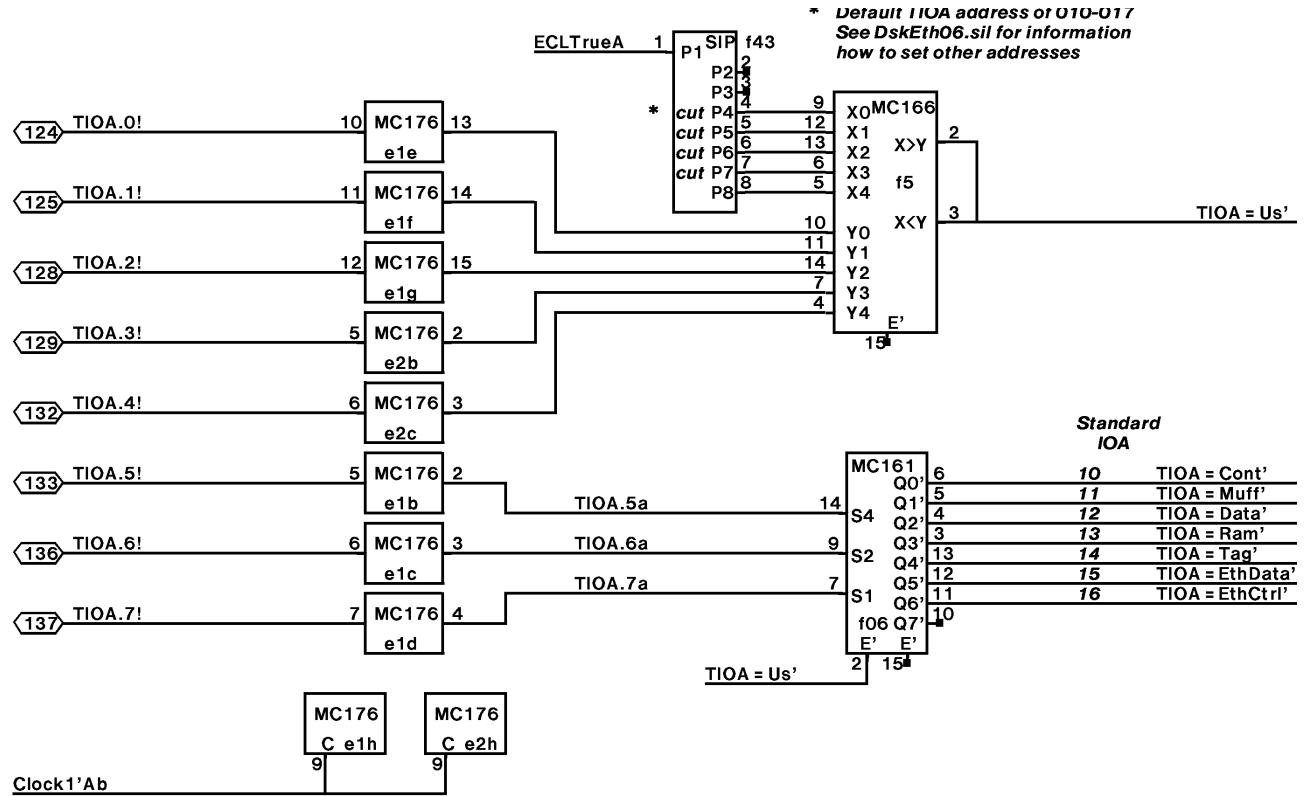
Character w/ code 124b, font Helvetica7BI would lie off the page.

Character w/ code 105b, font Helvetica7BI would lie off the page.

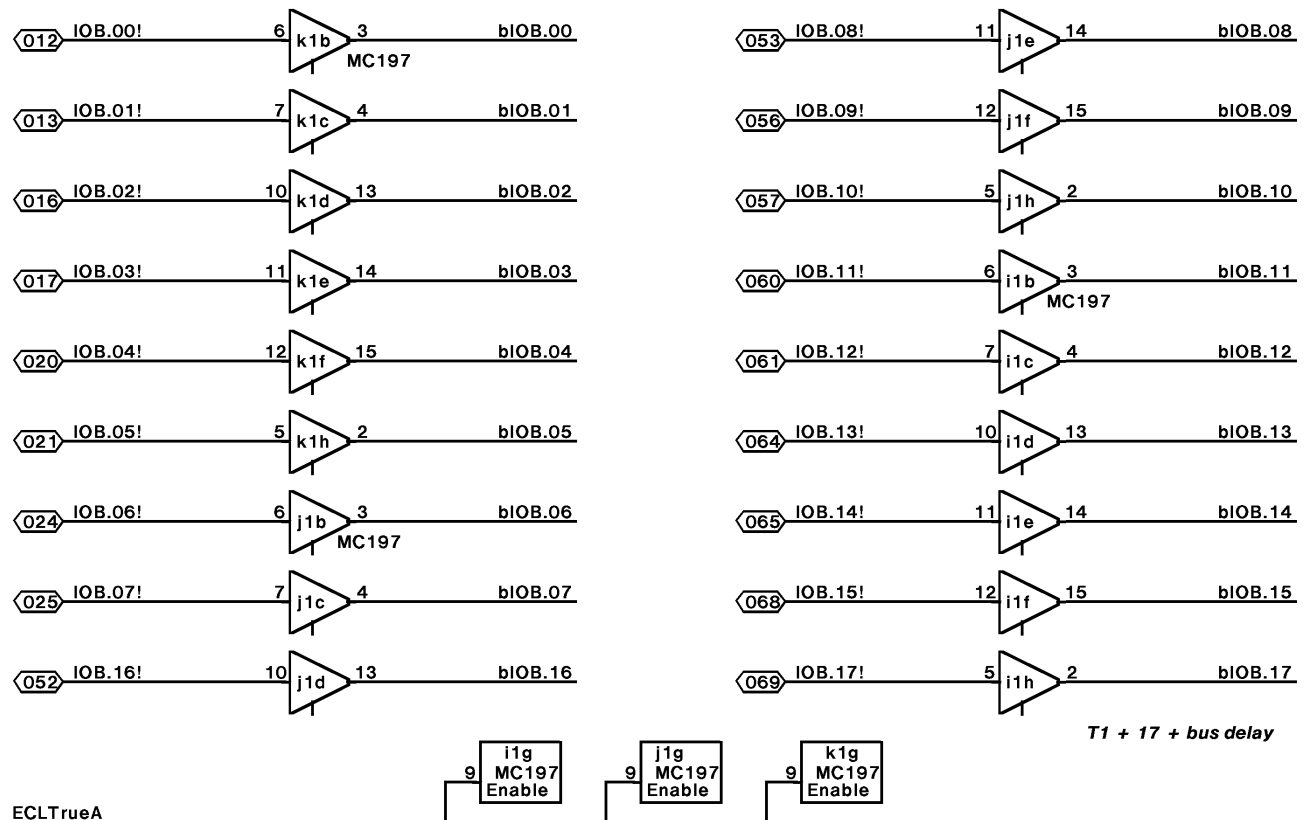


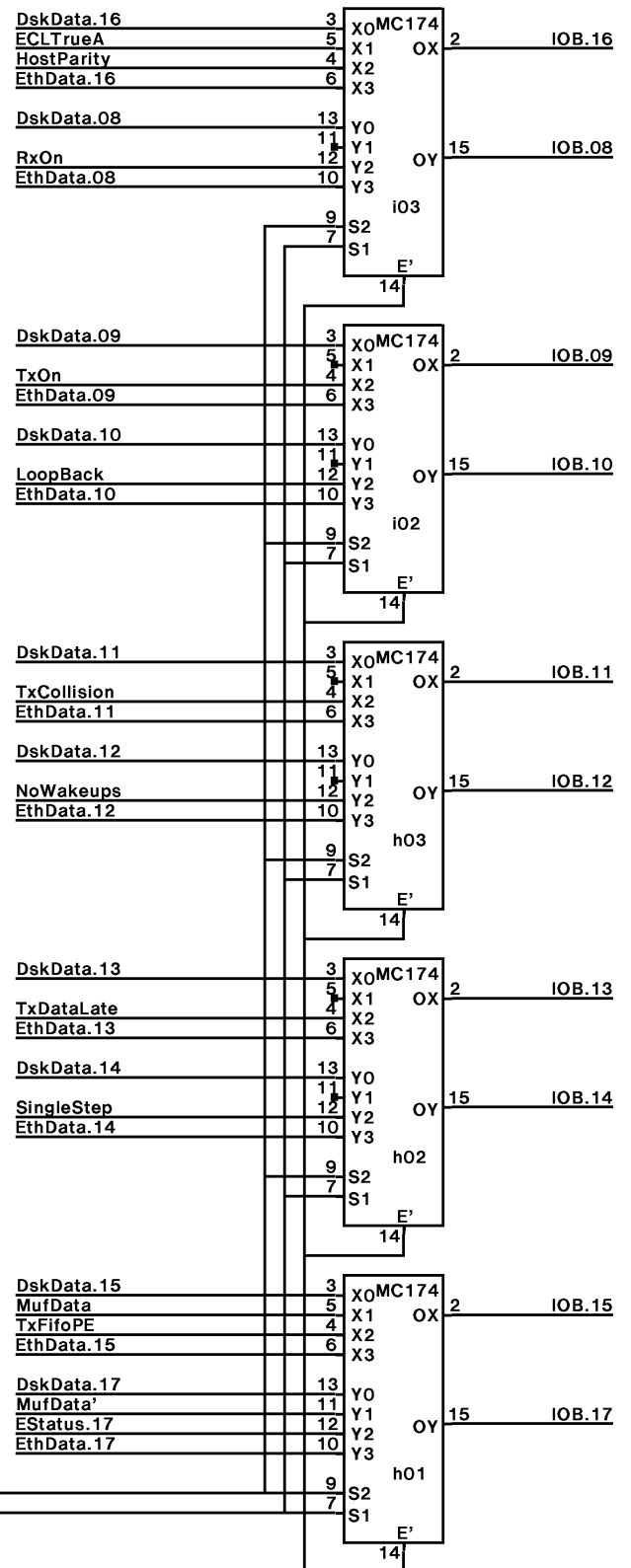
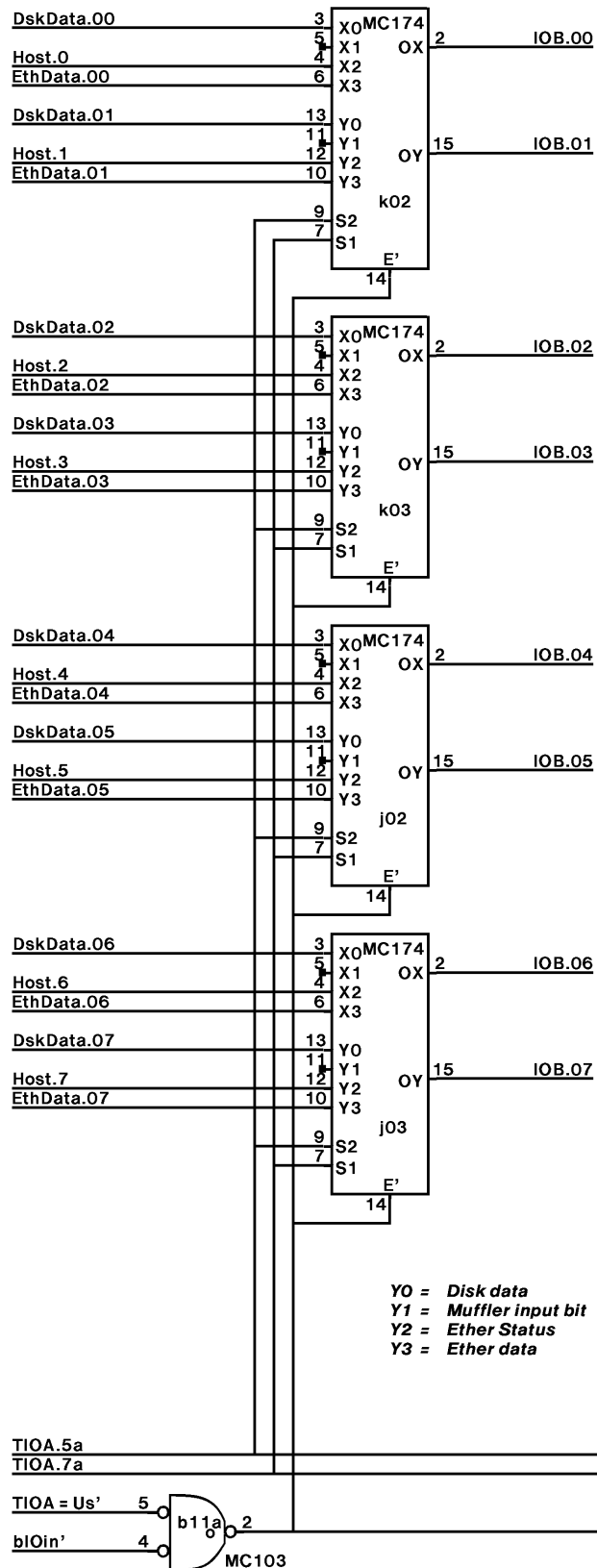
1) Drawings of common logic	01
Midas Muffler Control	01
TIOA and IOB receivers/drivers	02
Common Clock and Temperature Sense	04
Layout	05
2) Drawings for TriconD disk Controller	07
State Control Register	08
Format Ram, Counter and Proms	09
Tag Register	10
Disk Drive Control	11
FIFO	13
Error Correction Shift Register	15
Task Wake-Up and IOB parity check	16
Disk Mufflers	18
Dorado and Disk Clock Distribution	19
I/O pins and Termination	20
Timing Diagram	21
Cable Assembly Drawings	22
3) Drawings for Ethernet Controller	23
Phase Decoder	24
Receiver	25
Transmitter	29
Phase Encoder	33
Test Logic	34
Dorado and Ether Clock Distribution	35
Next Bus and IOattention	36
Ether Mufflers	37
Cable I/O and termination	38
Timing Diagrams	40

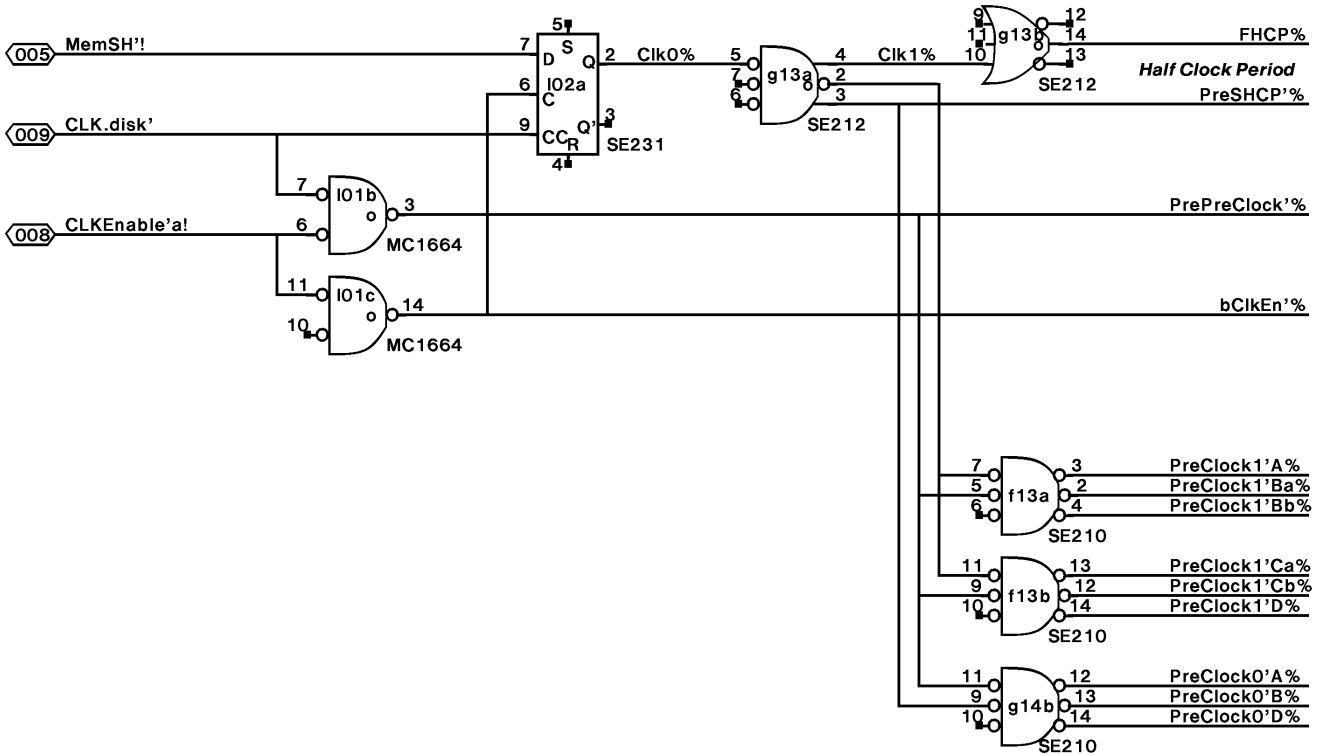




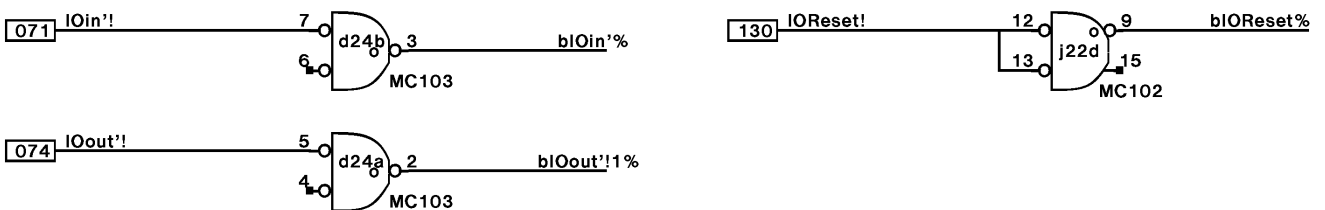
IOB receivers



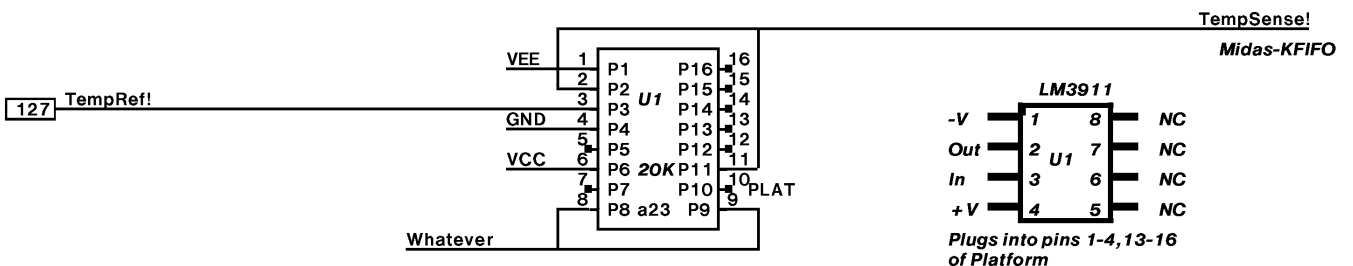




Common Clock circuitry



Temperature Sensor

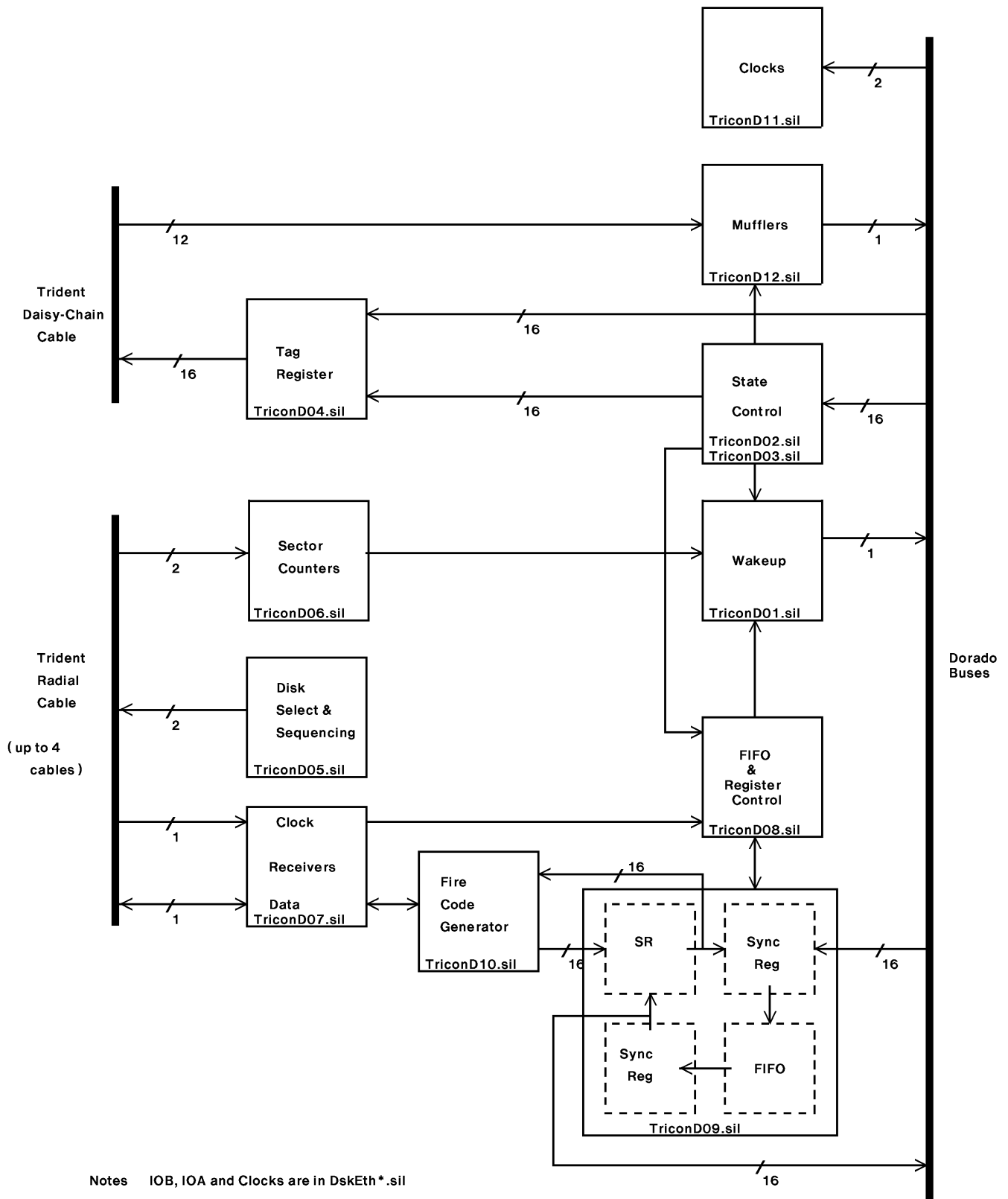


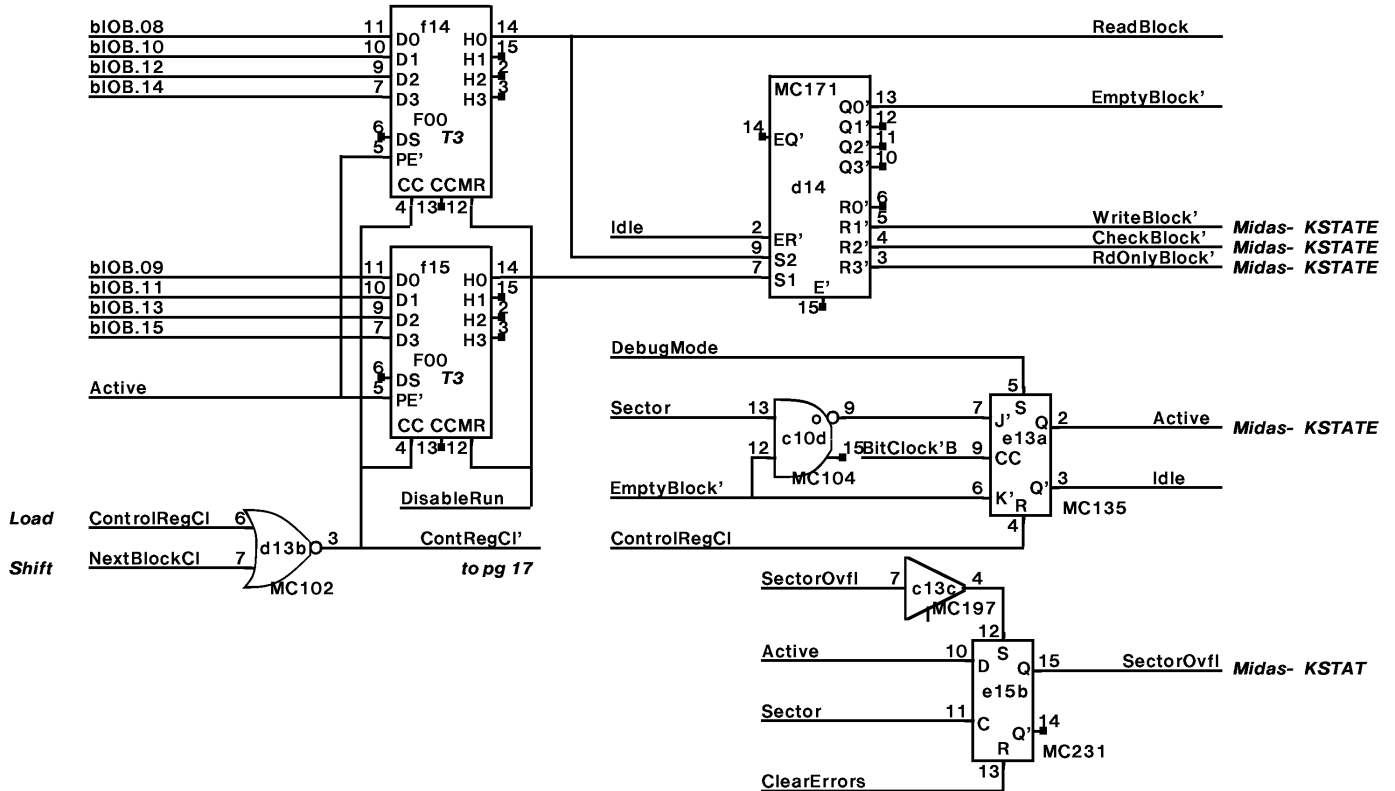
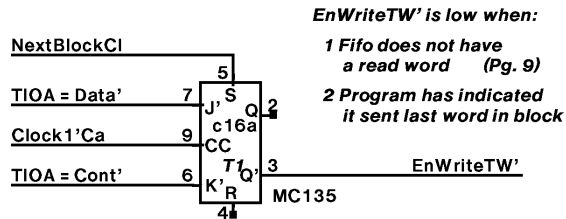
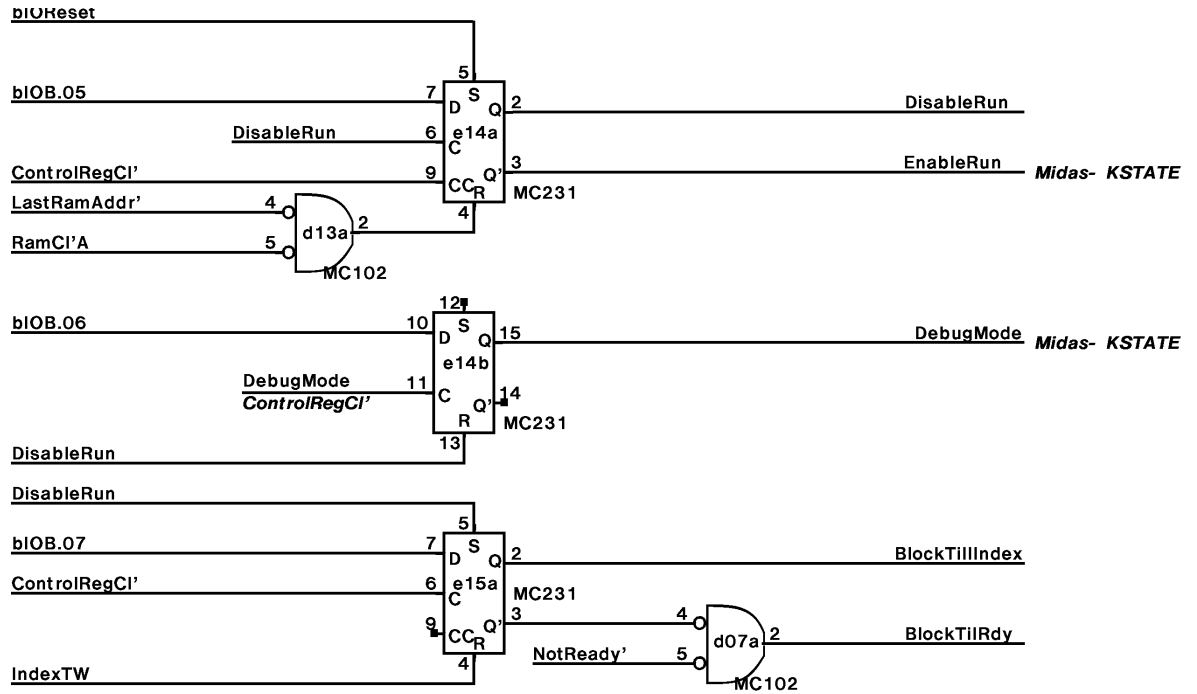
Cut SIP legs at k52 to set the
Muffler addresses for the board

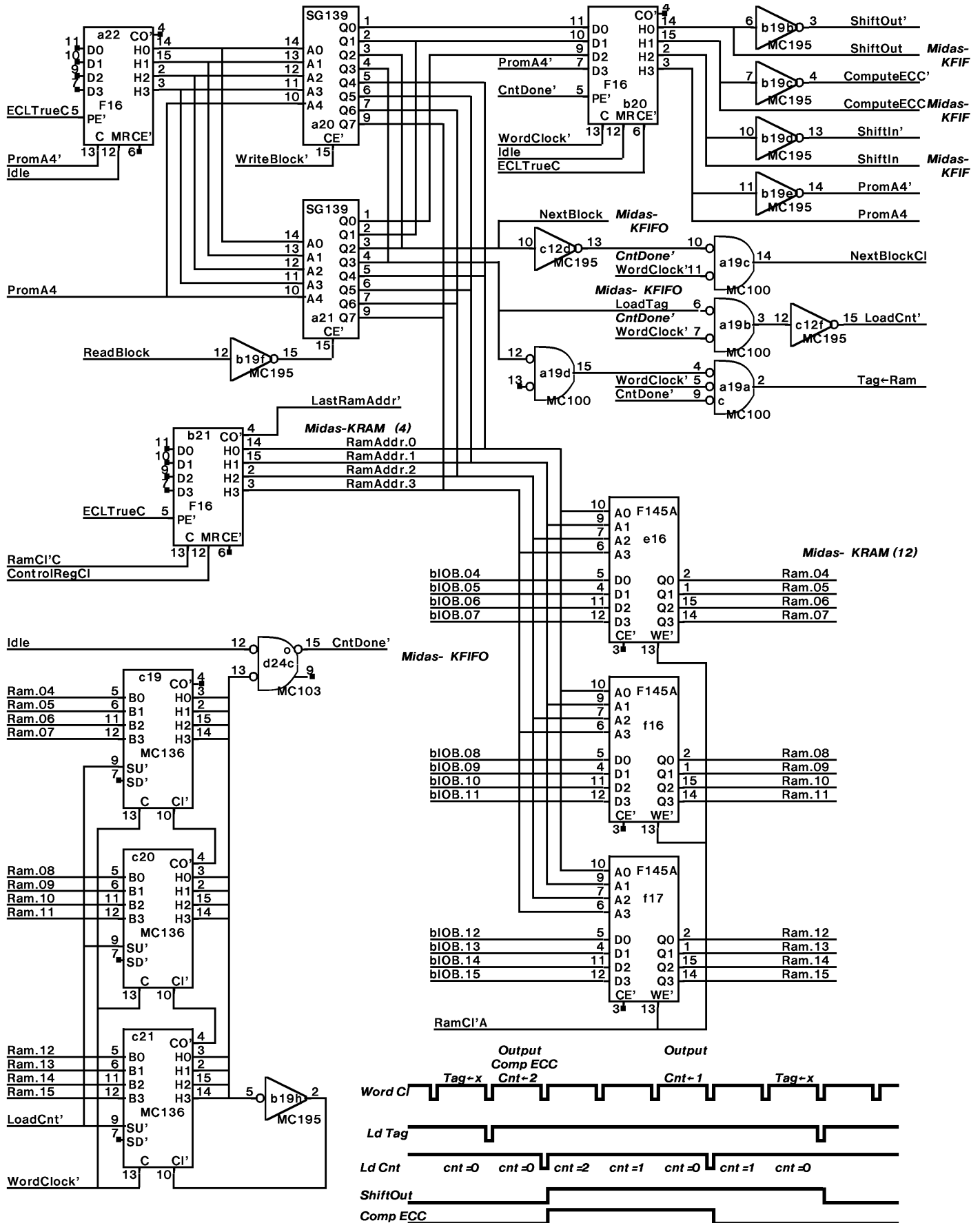
Muff Addr	P5	P6	P7	P8
0000-0177	cut	cut	cut	cut
0200-0377	cut	cut	cut	
0400-0577	cut	cut		cut
0600-0777	cut	cut		
1000-1177	cut		cut	cut
1200-1377	cut		cut	
1400-1577	cut			cut
1600-1777	cut			
2000-2177		cut	cut	cut
2200-2377		cut	cut	
2400-2577		cut		cut
2600-2777		cut		
3000-3177			cut	cut
3200-3377			cut	
3400-3577				cut
3600-3777				

Cut SIP legs at f43 to set the
IOA bus addresses for the board

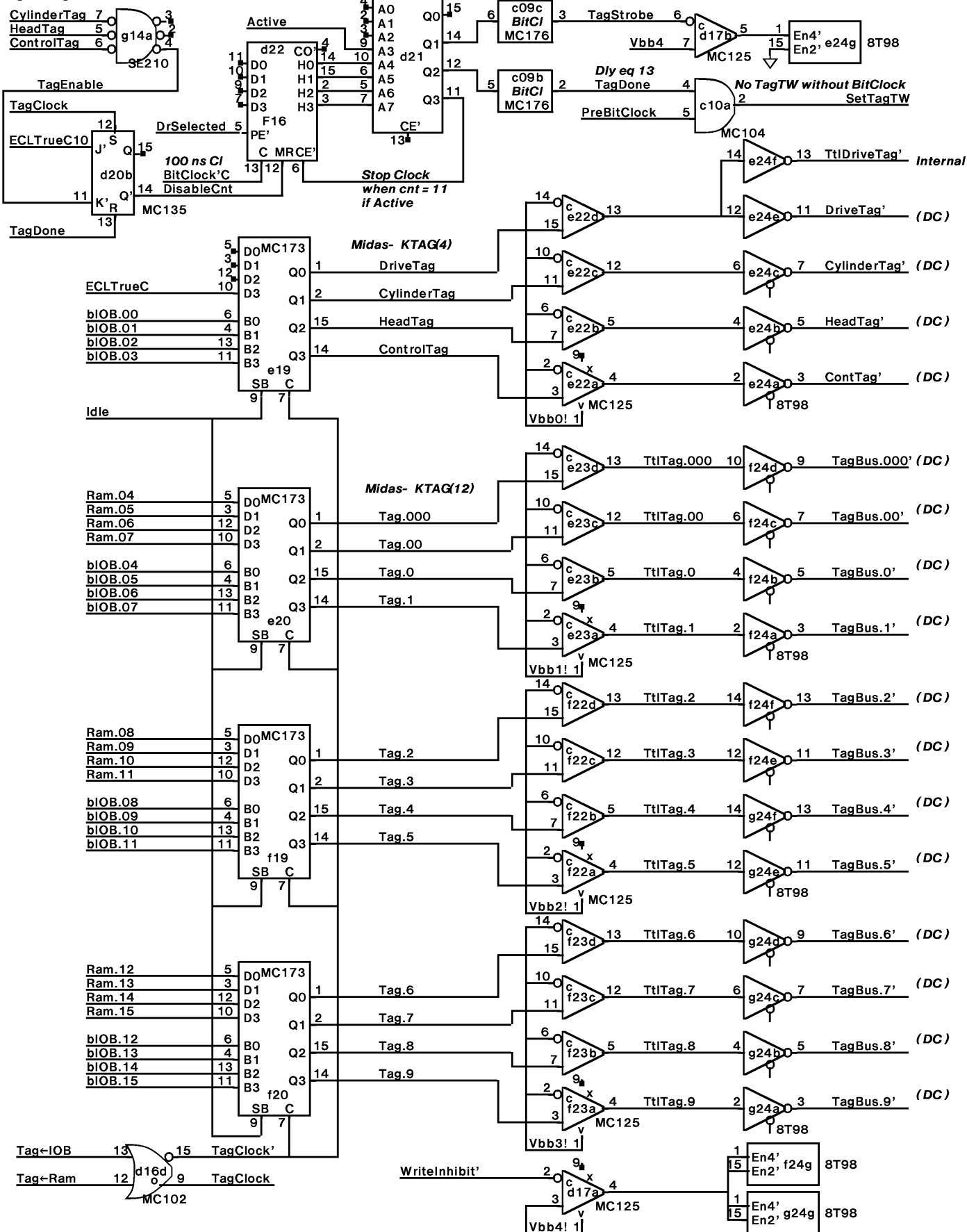
IOA	P4	P5	P6	P7	P8
000-007	cut	cut	cut	cut	cut
010-017	cut	cut	cut	cut	
020-027	cut	cut	cut		cut
030-037	cut	cut	cut		
040-047	cut	cut		cut	cut
050-057	cut	cut		cut	
060-067	cut	cut			cut
070-077	cut	cut			
100-107	cut		cut	cut	cut
110-117	cut		cut	cut	
120-127	cut		cut		cut
130-137	cut		cut		
140-147	cut			cut	cut
150-157	cut			cut	
160-167	cut				cut
170-177	cut				
200-207		cut	cut	cut	cut
210-217		cut	cut	cut	
220-227		cut	cut		cut
230-237		cut	cut	cut	
240-247		cut		cut	cut
250-257		cut			
260-267		cut			cut
270-277		cut			
300-307			cut	cut	cut
310-317			cut	cut	
320-327			cut		cut
330-337			cut		
340-347				cut	cut
350-357				cut	
360-367					cut
370-377					



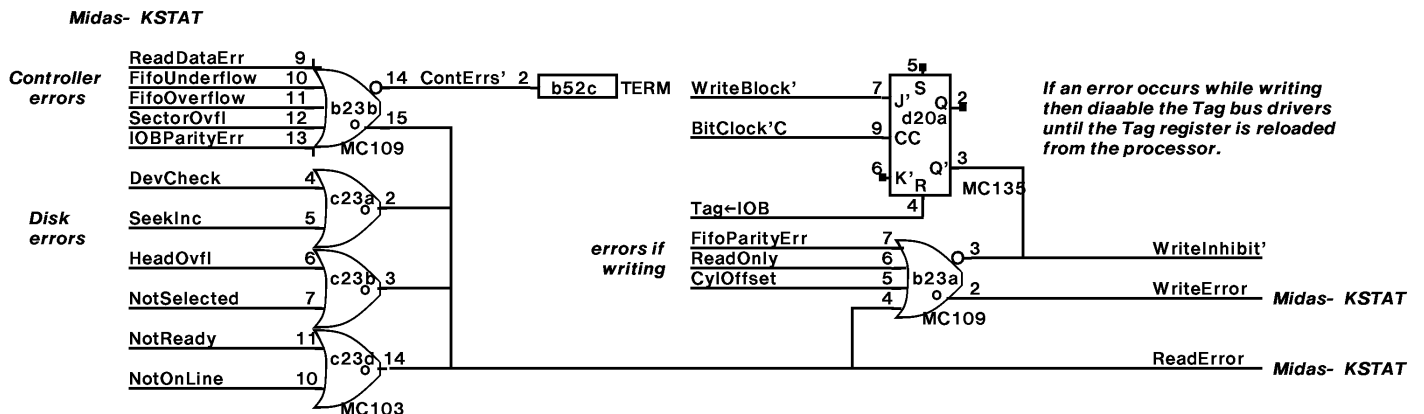
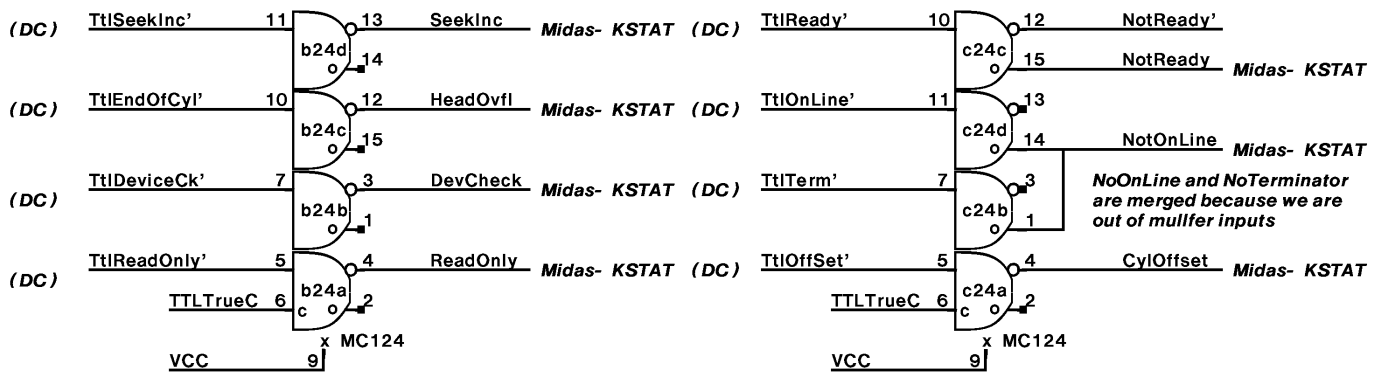
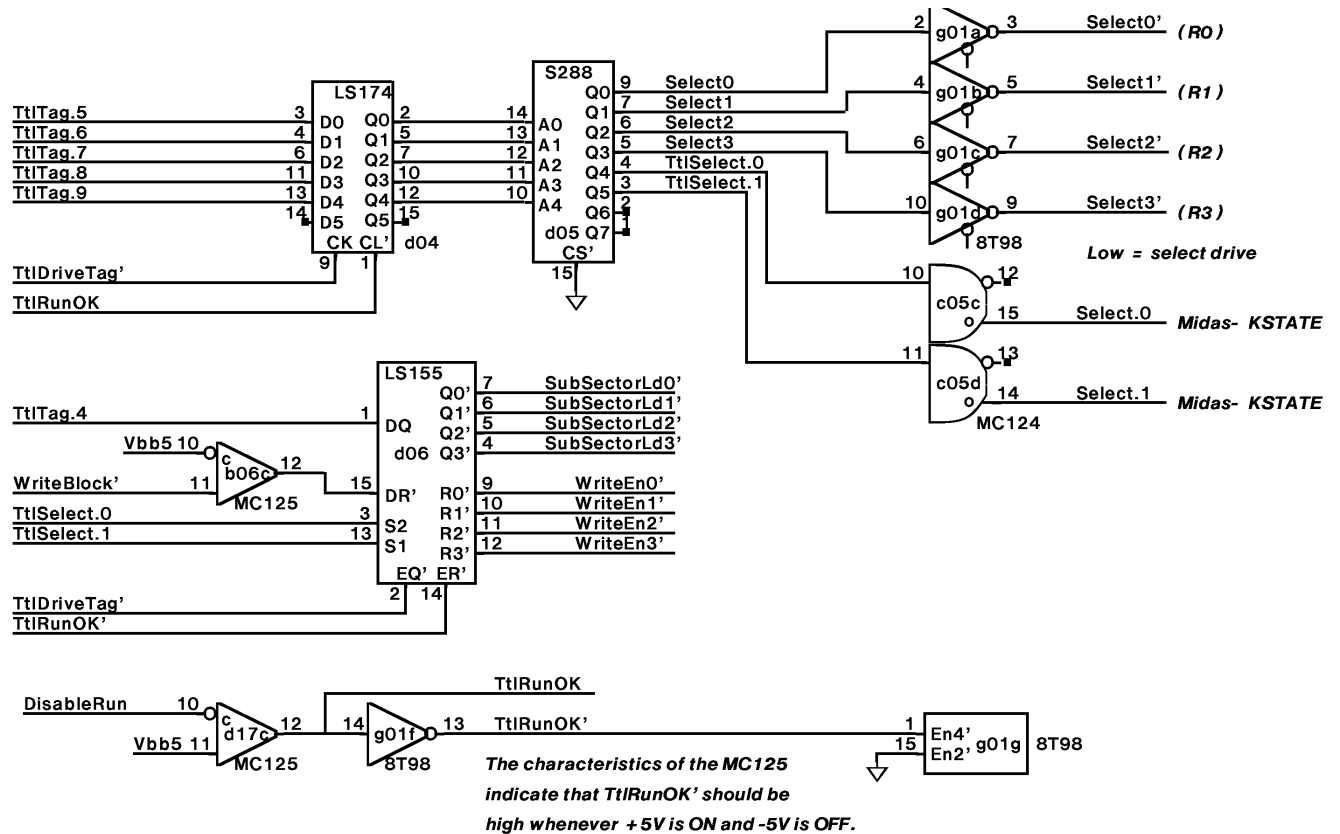


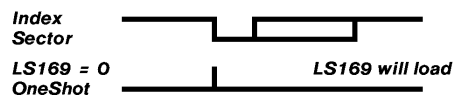
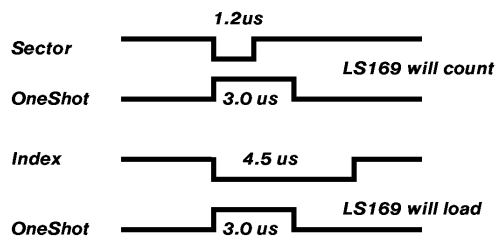
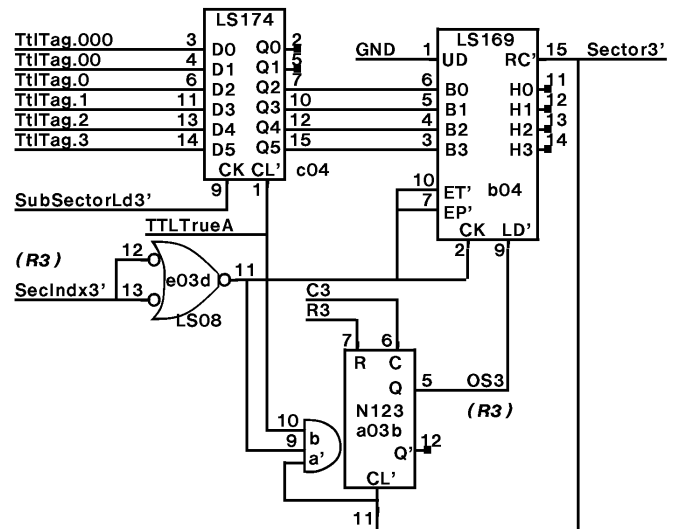
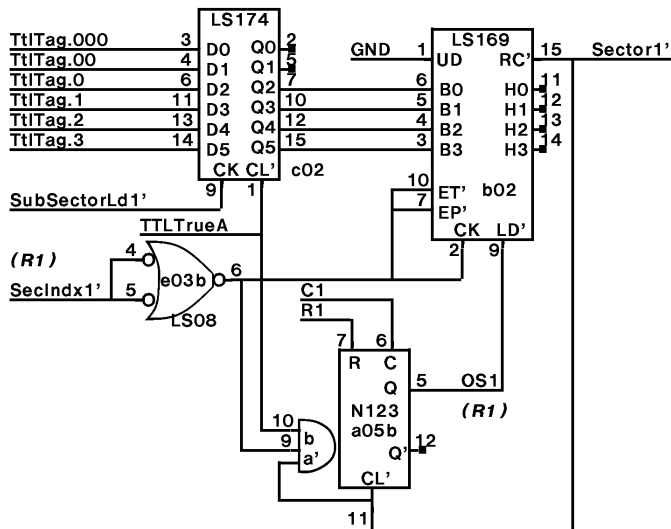
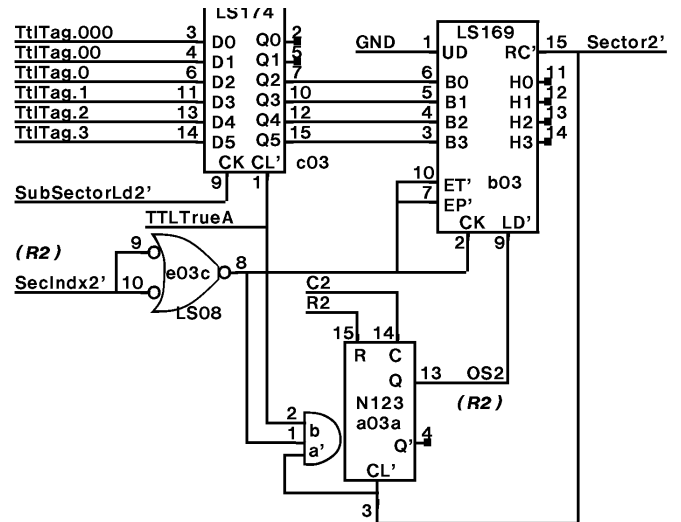
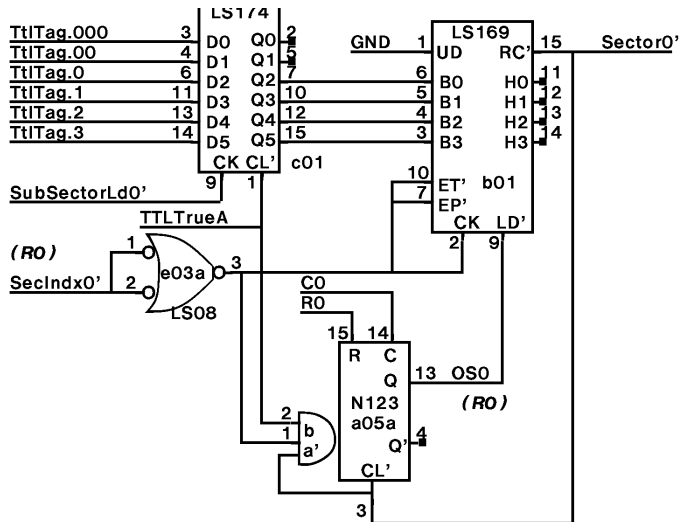


No Tag Timing if not one of:



XEROX	Project	Drawing	File	Designer	Rev	Date	Page
PARC	Dorado	TAG register & Bus Drivers	TriconD04.sil	Roger Bates	Bd	1/22/79	10



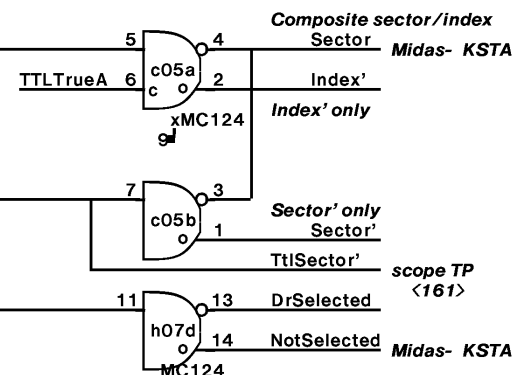
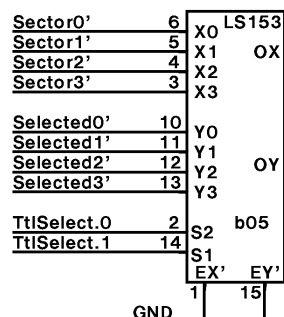


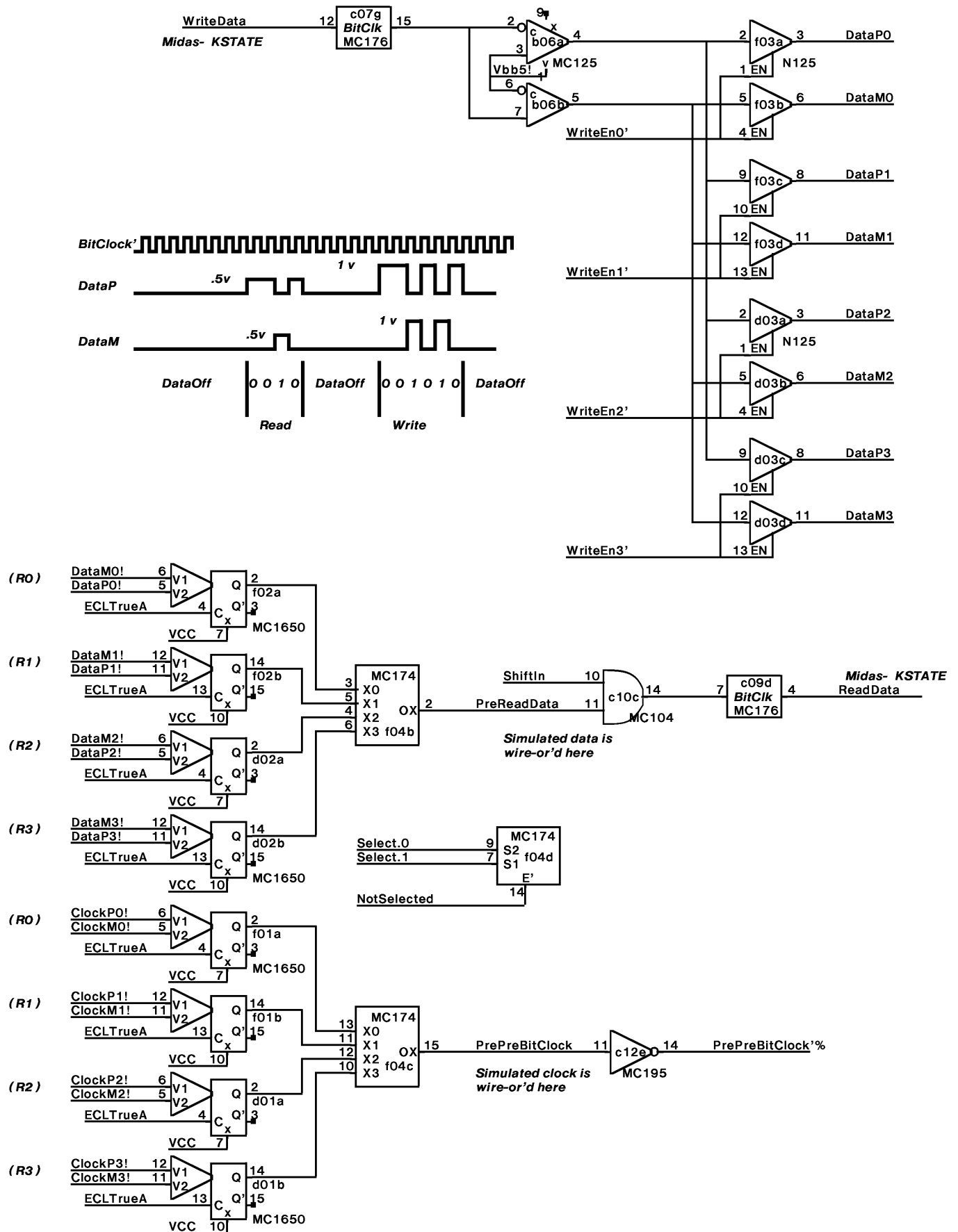
R0	1	P1 27k	P16	VCC
	2	P2 330pf	P15	C0
R1	3	P3 27k	P14	
	4	P4 330pf	P13	C1
R2	5	P5 27k	P12	
	6	P6 330pf	P11	C2
R3	7	P7 27k	P10	
	8	P8 330pf	P9	C3
		PLAT	a04	

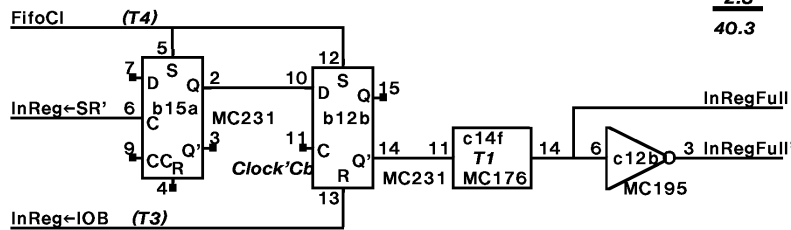
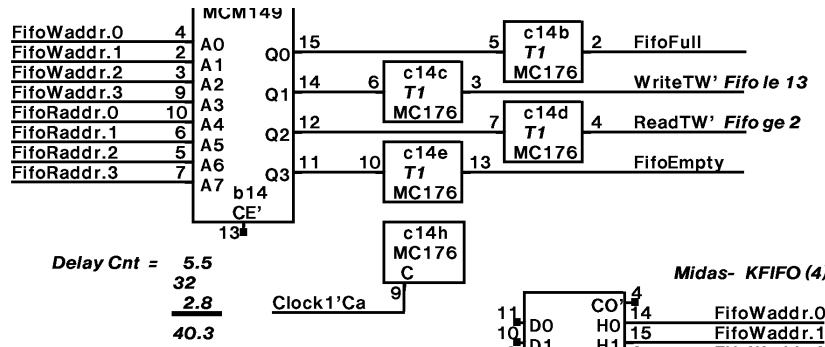
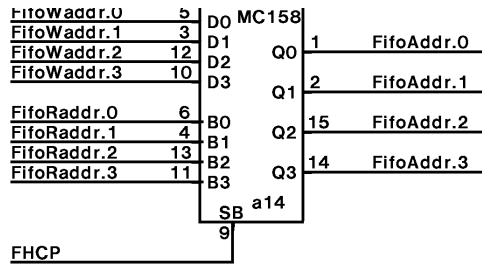
The above sector counters require the TRIDENT disk to have its sector counters set to provide 117 "sub-sector" pulses per revolution. This is done by setting the disk jumpers as follows:

X6A	X6B
4 - 11	2 - 13
5 - 10	3 - 12
	4 - 11
	5 - 10
	6 - 09

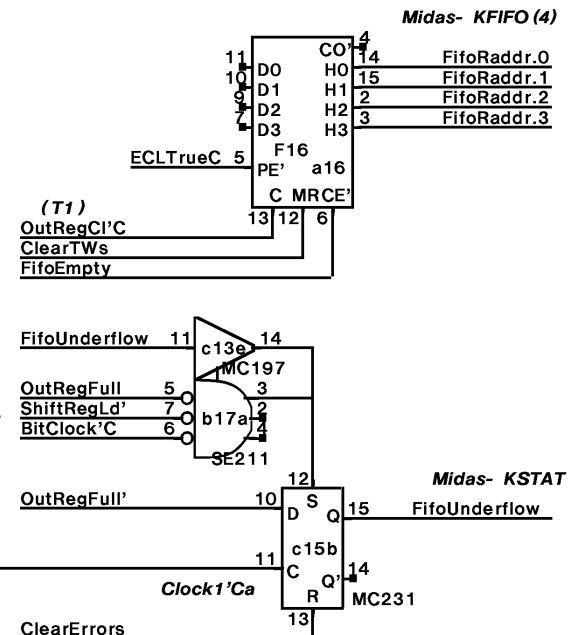
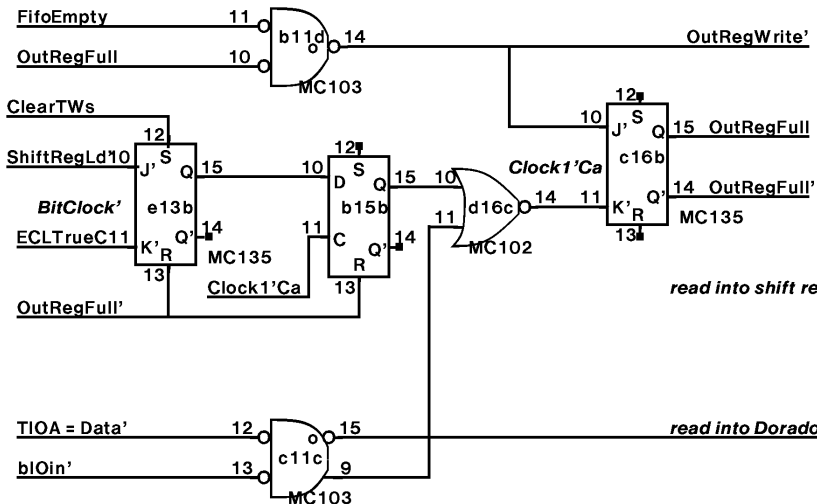
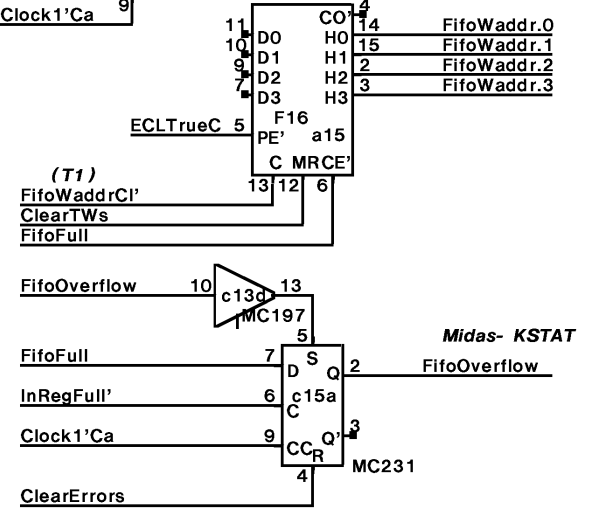
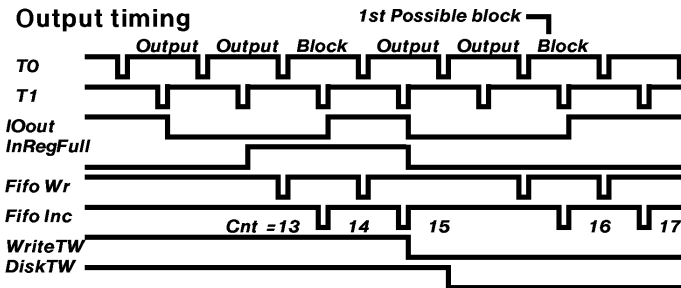
(R0)
 (R1)
 (R2)
 (R3)



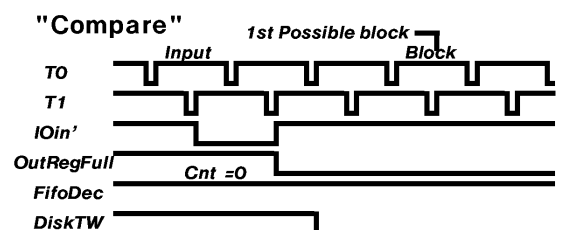
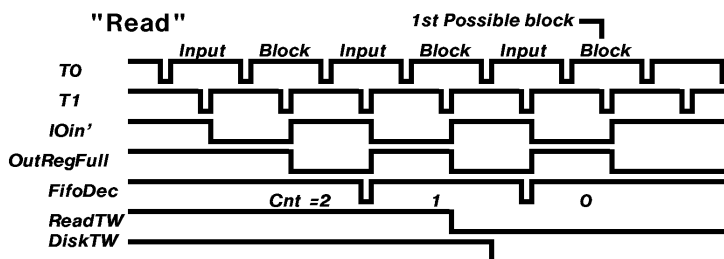




Output timing

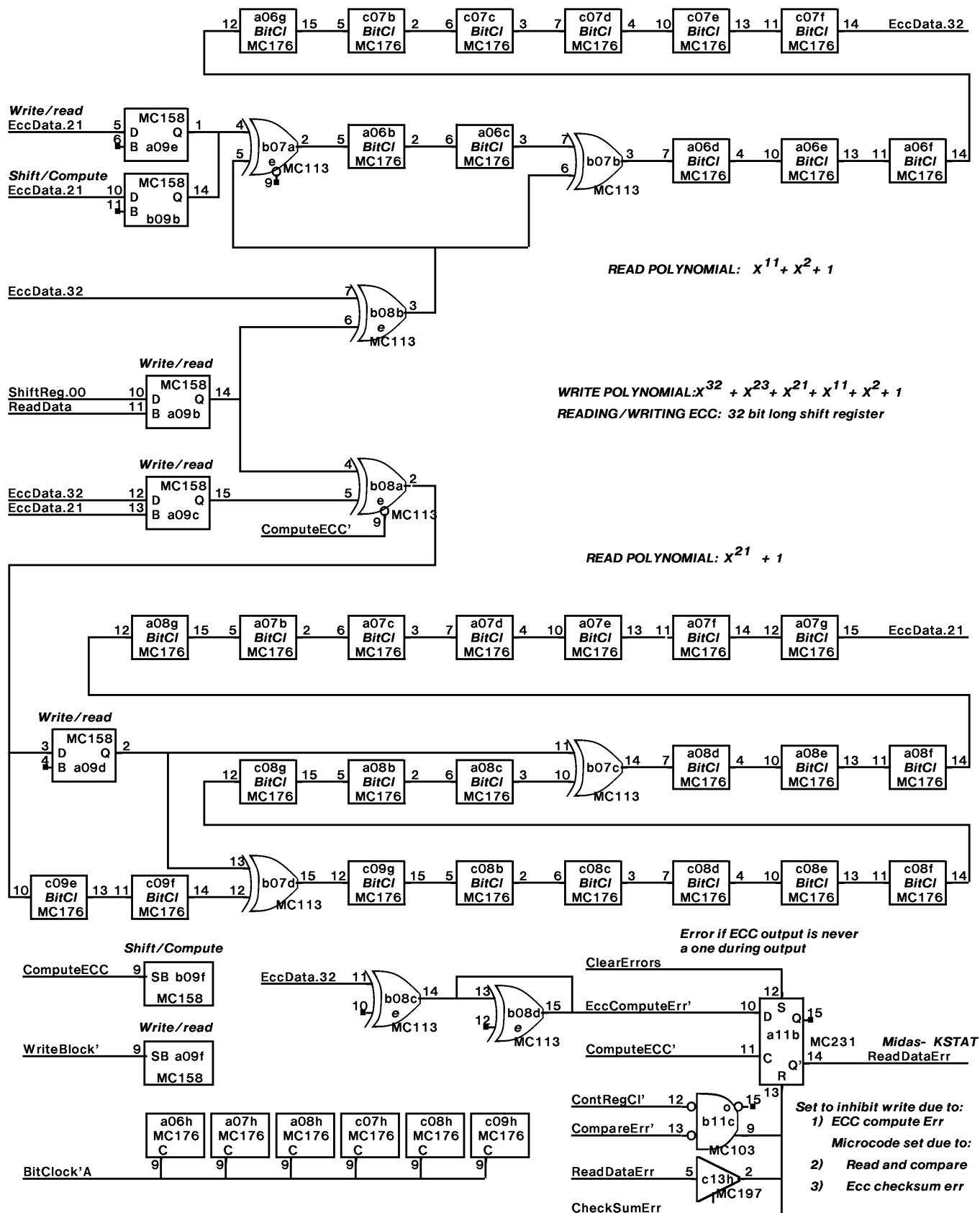


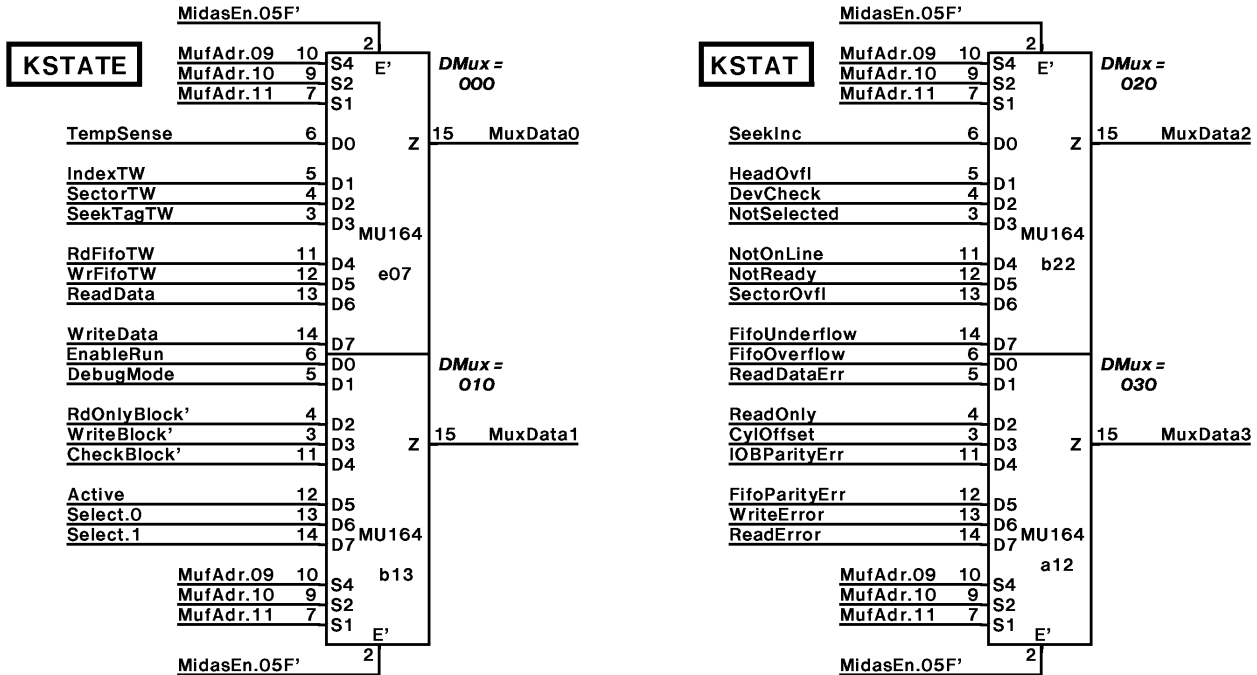
Input timing





POLYNOMIAL DIVIDER FOR FIRE CODE GENERATION

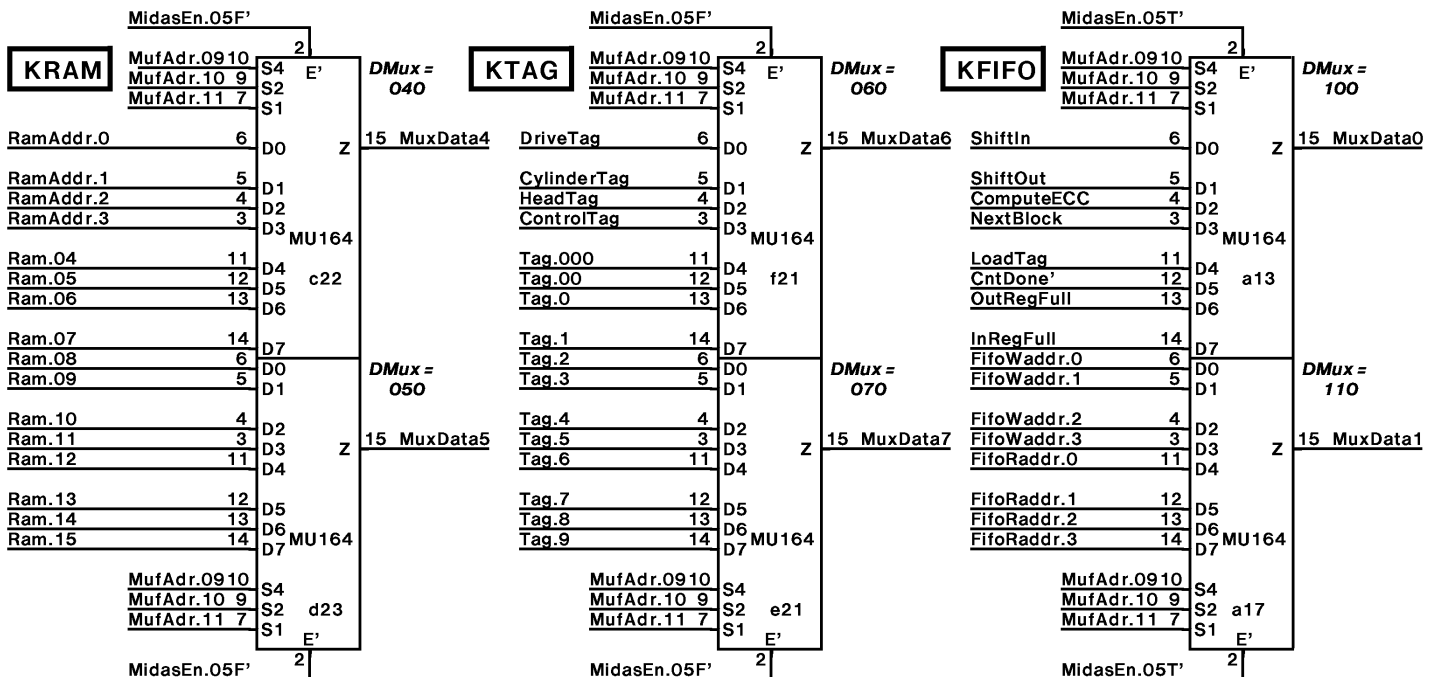


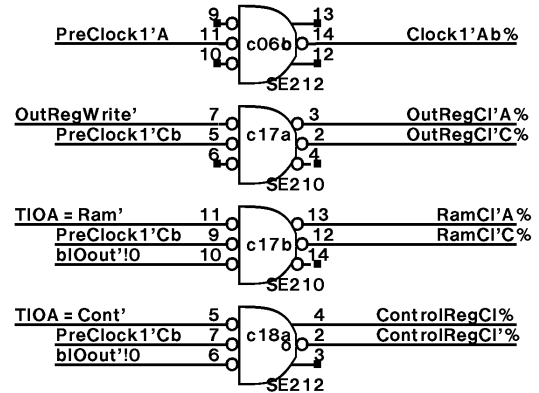
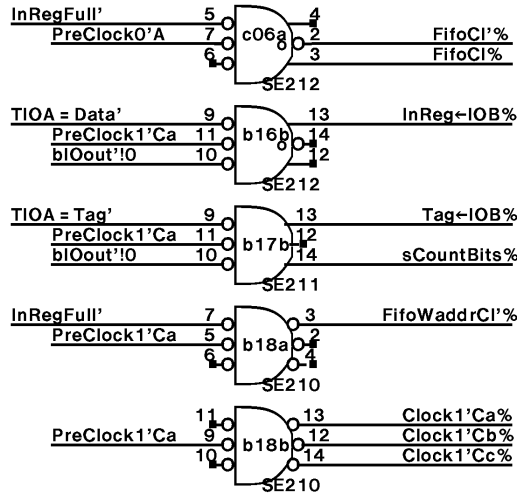
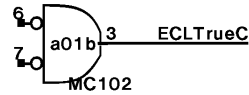
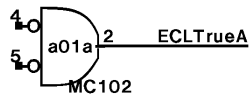


Muffler addresses are:

Value listed for Program input
Value plus 2000 for Midas input

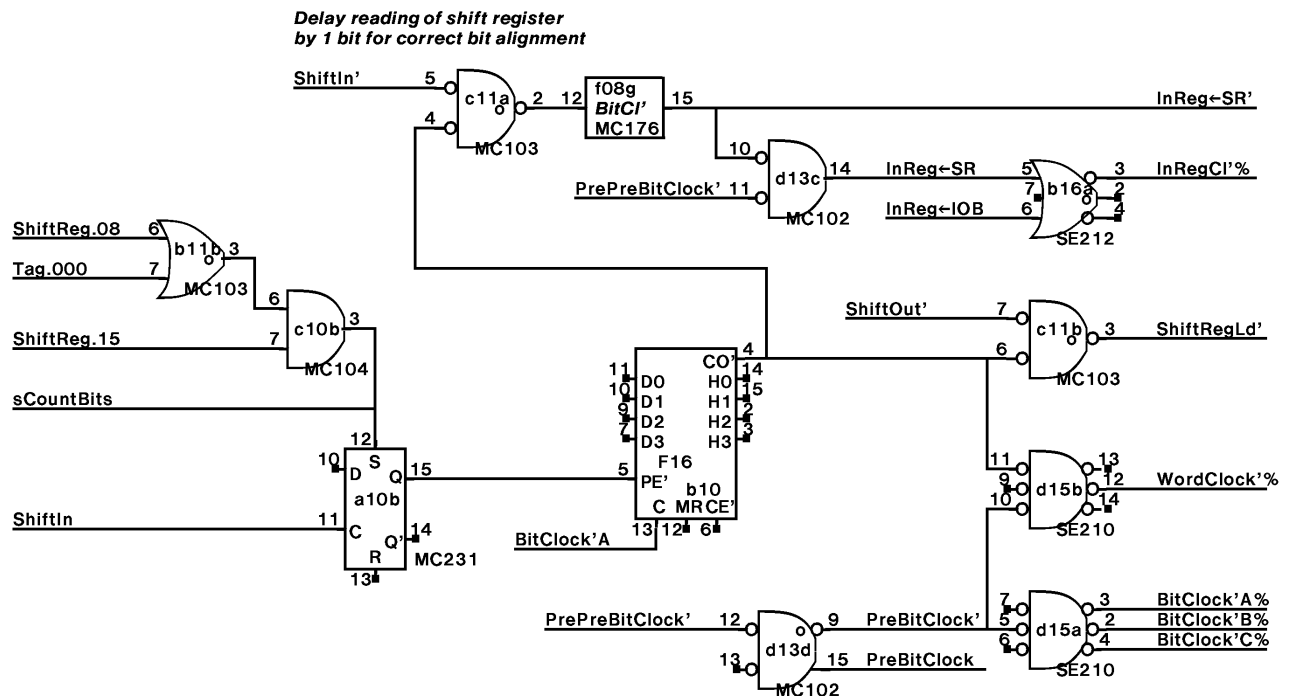
Values from 120 to 177 are
used by the Ethernet



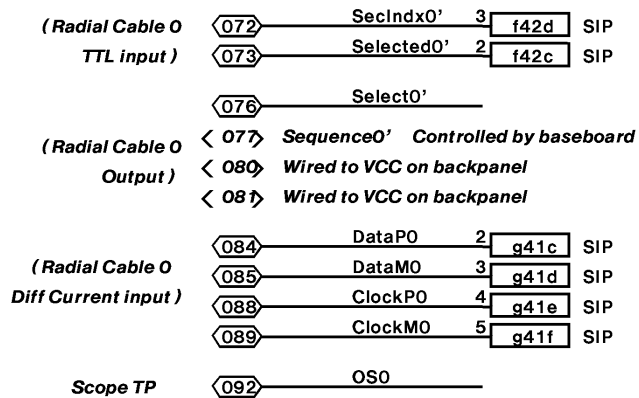


System Clocks

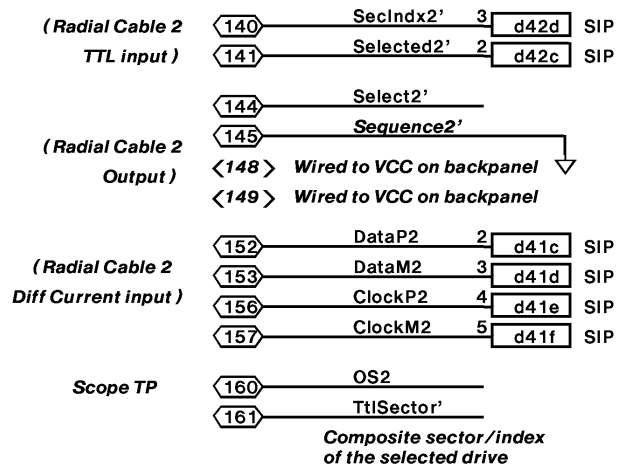
Disk Clocks



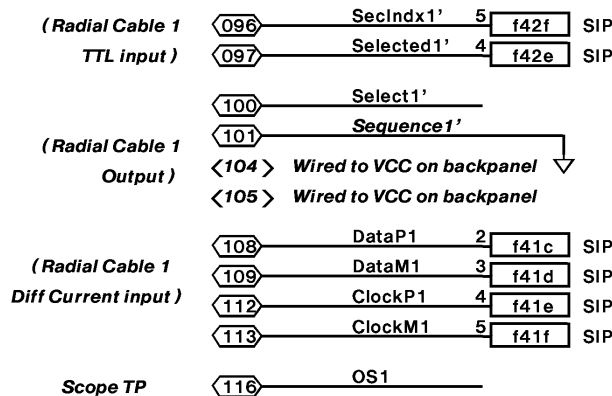
Radial Cable for Drive 0



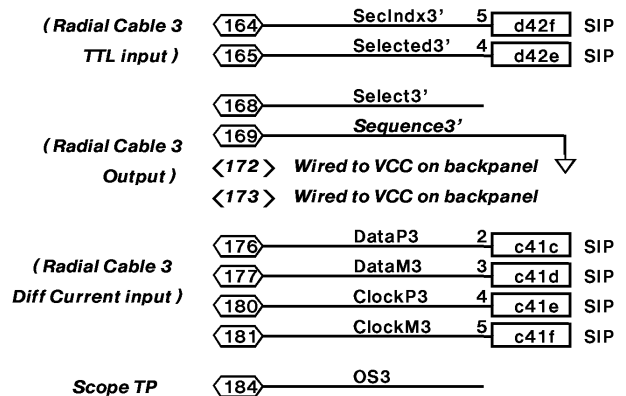
Radial Cable for Drive 2



Radial Cable for Drive 1

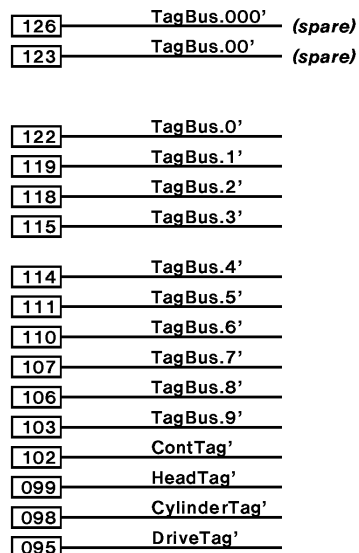


Radial Cable for Drive 3

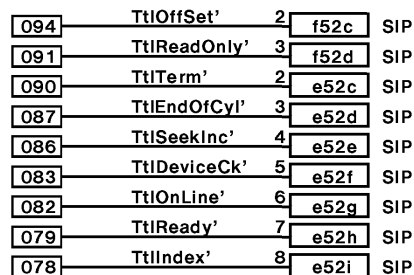


Daisy Chain Cable

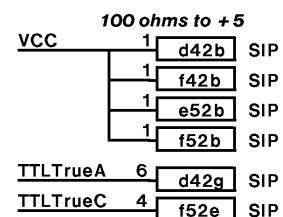
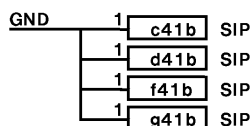
Daisy chain outputs



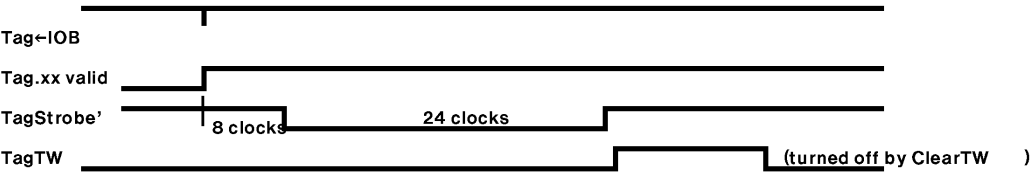
Daisy chain TTL inputs



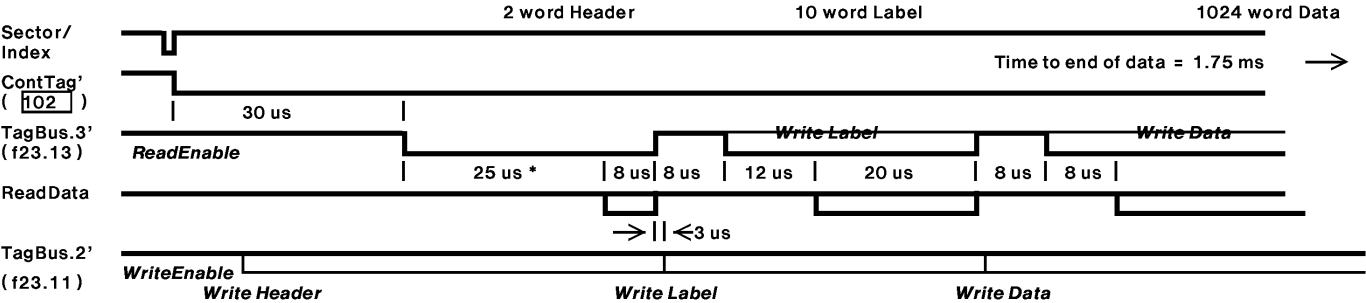
100 ohms to ground



Head or Cylinder Tag Instruction



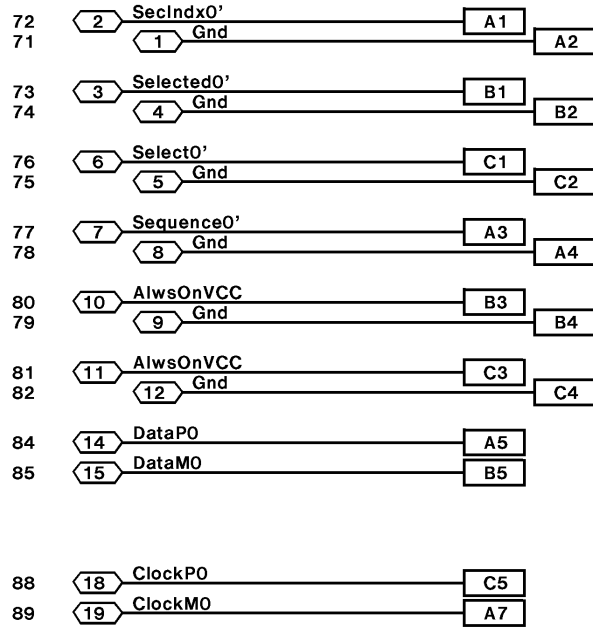
Read or Write Instruction



* This value is for reading a pack on the same drive that Headers were written
It may vary by +/- 15 us on other drives

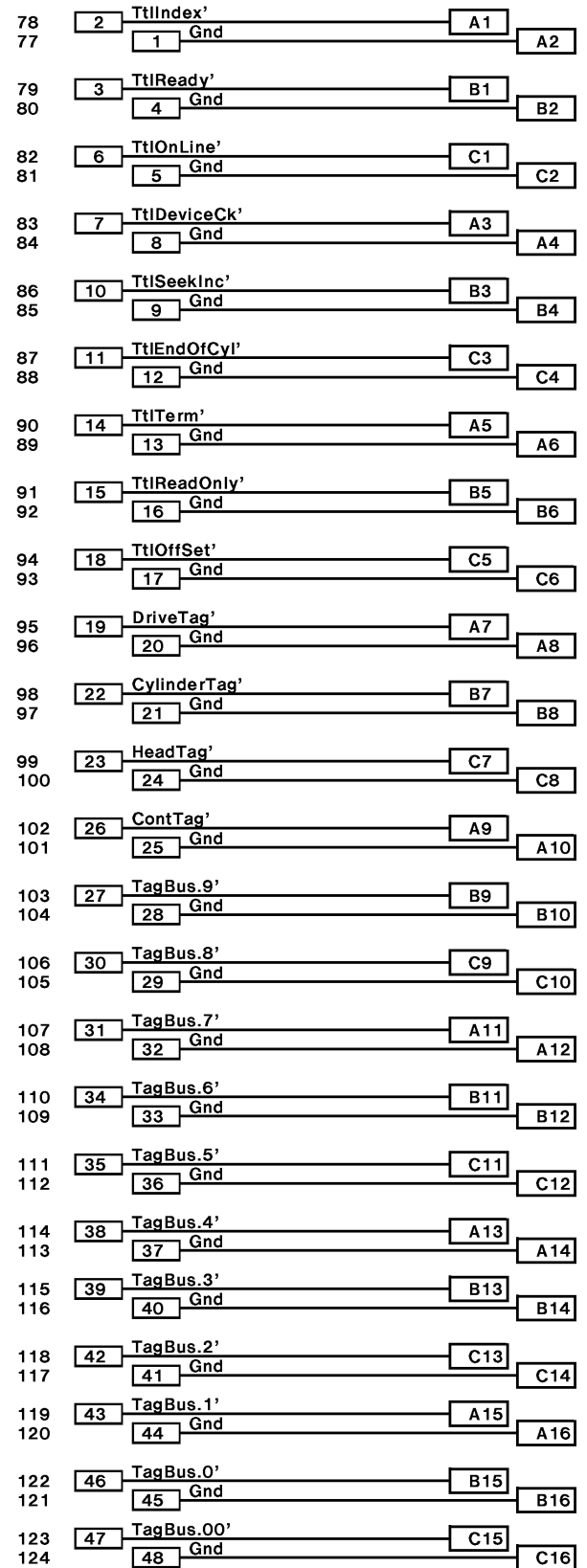
Radial Cable for Drive 0

AMP 204729-1



DAISEY CHAIN CABLE

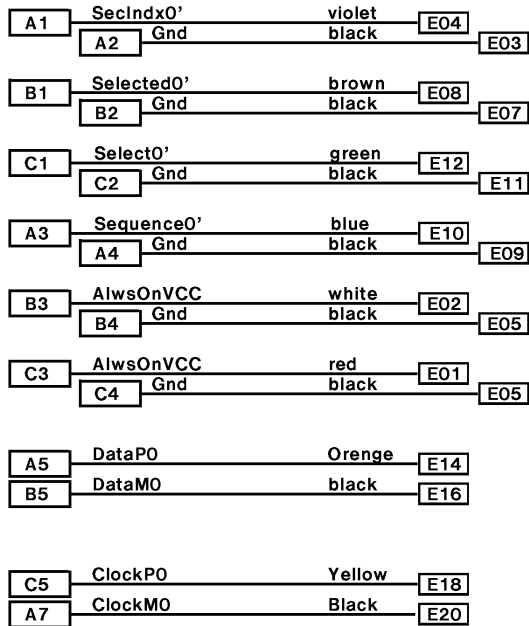
AMP 204733-1



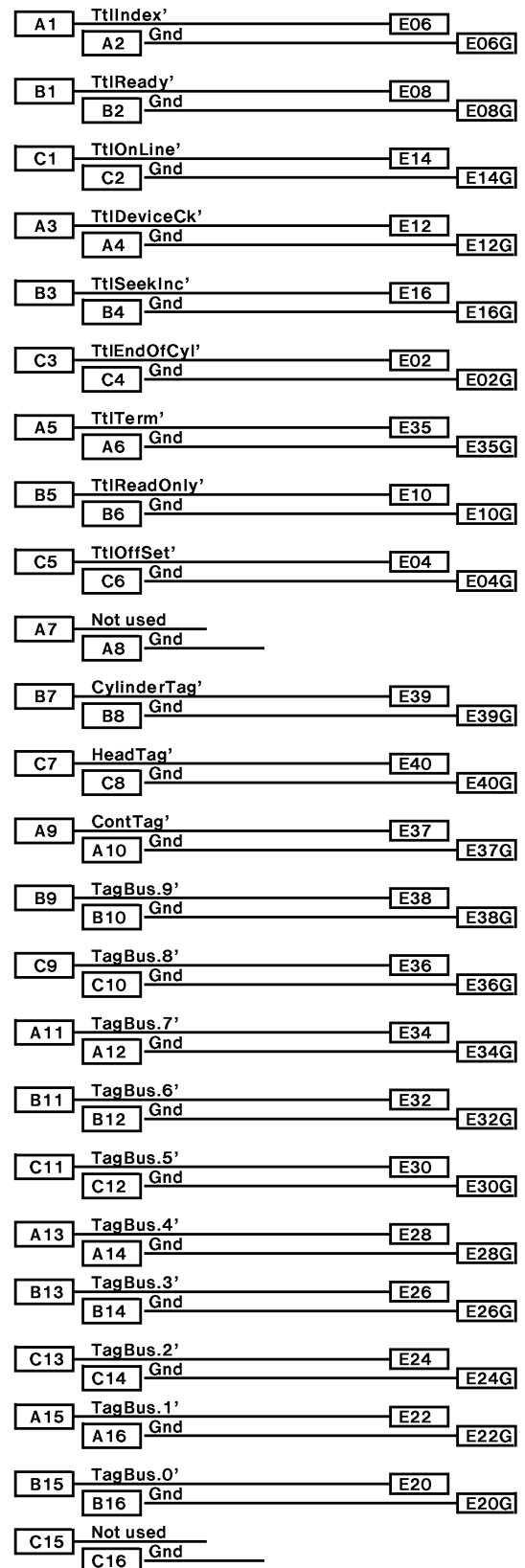
Radial Cable for Drive 0

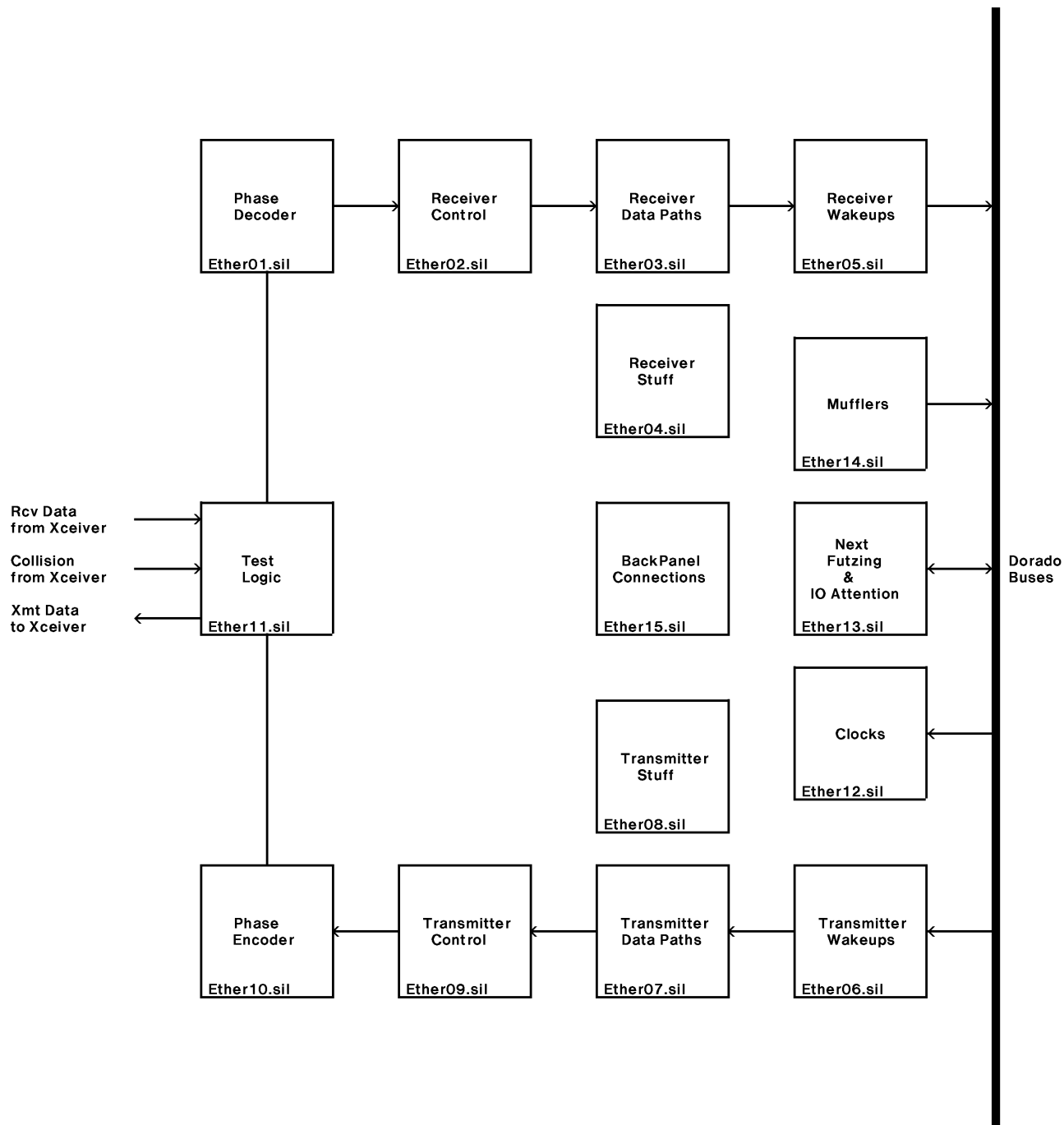
AMP 204742-1

Cal-Comp
Assembly 12433

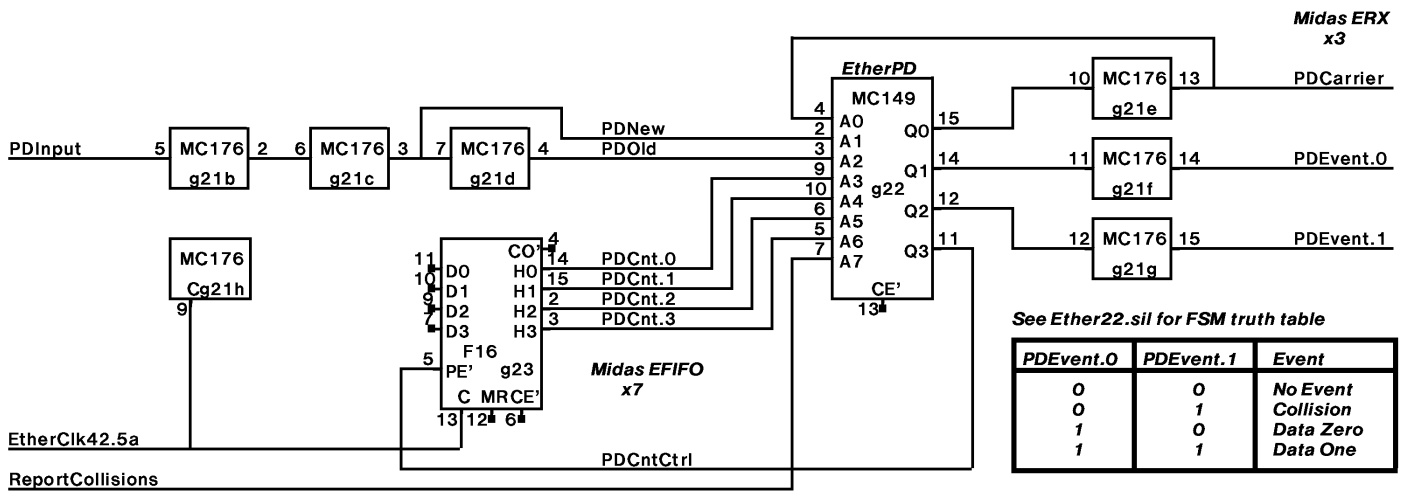


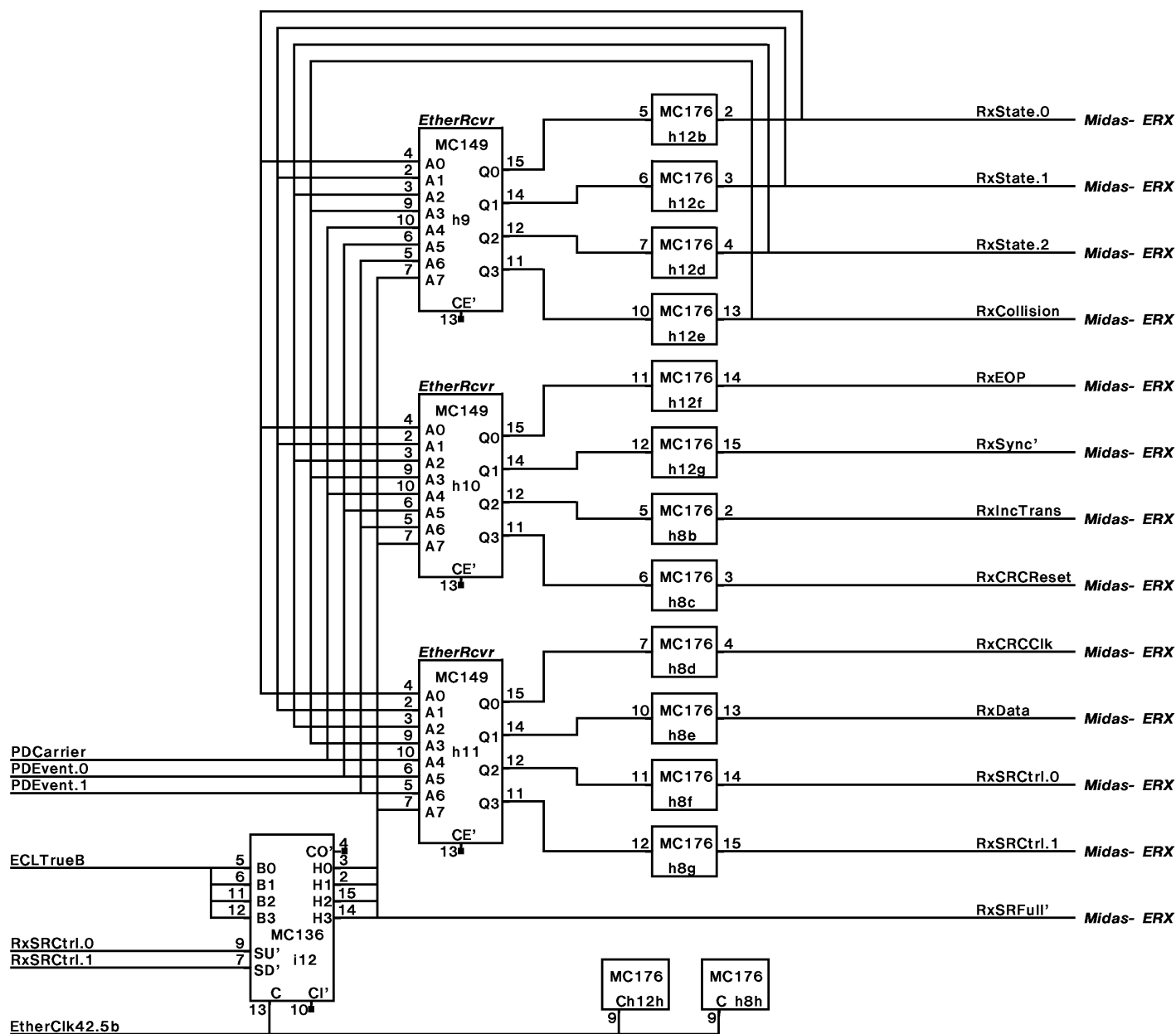
DAISEY CHAIN CABLE AMP 204746-1 Cal-Comp Assembly 12424





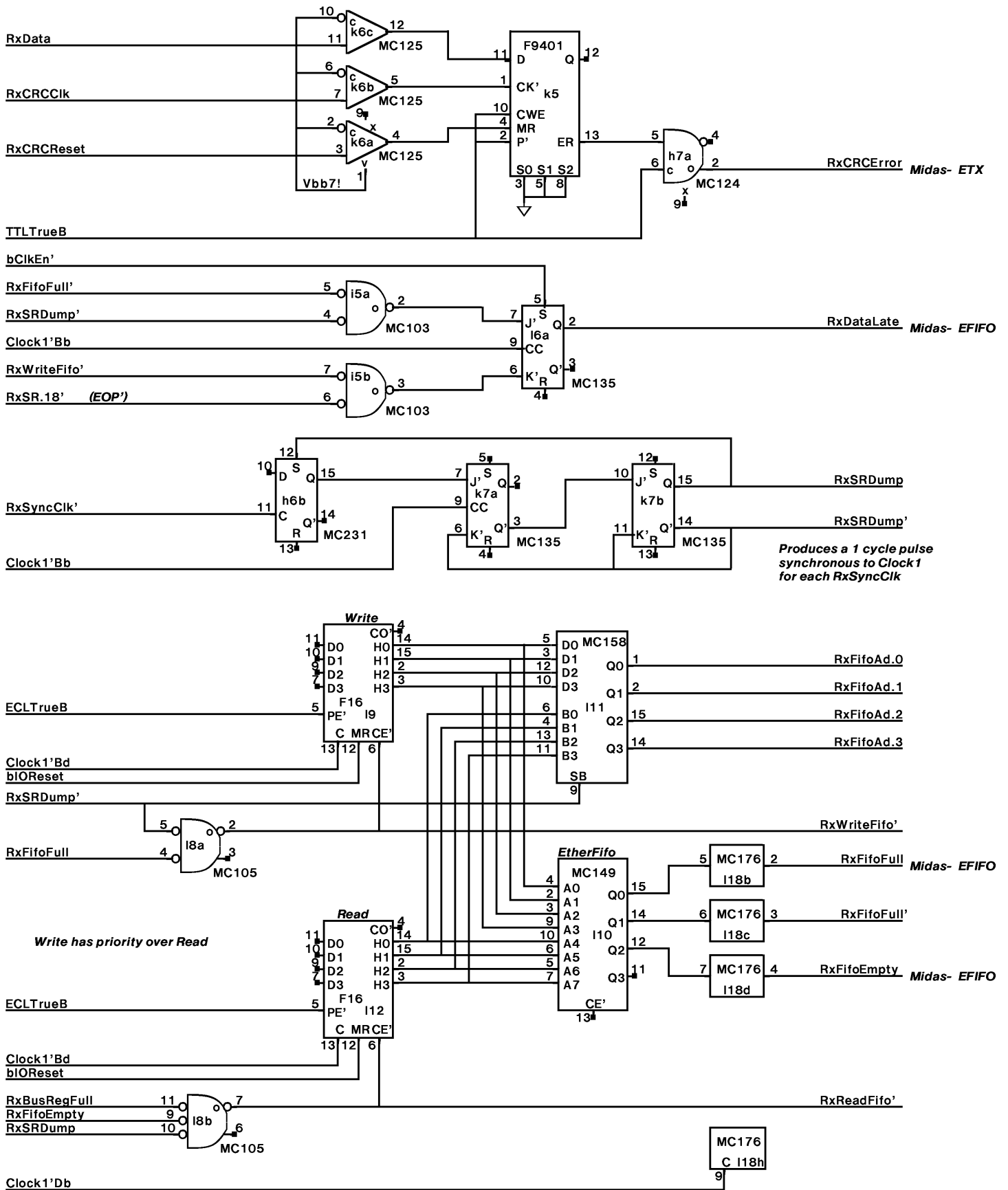
See DskEth*.sil for IOA, IOB, Muffler Control and Board Clocks

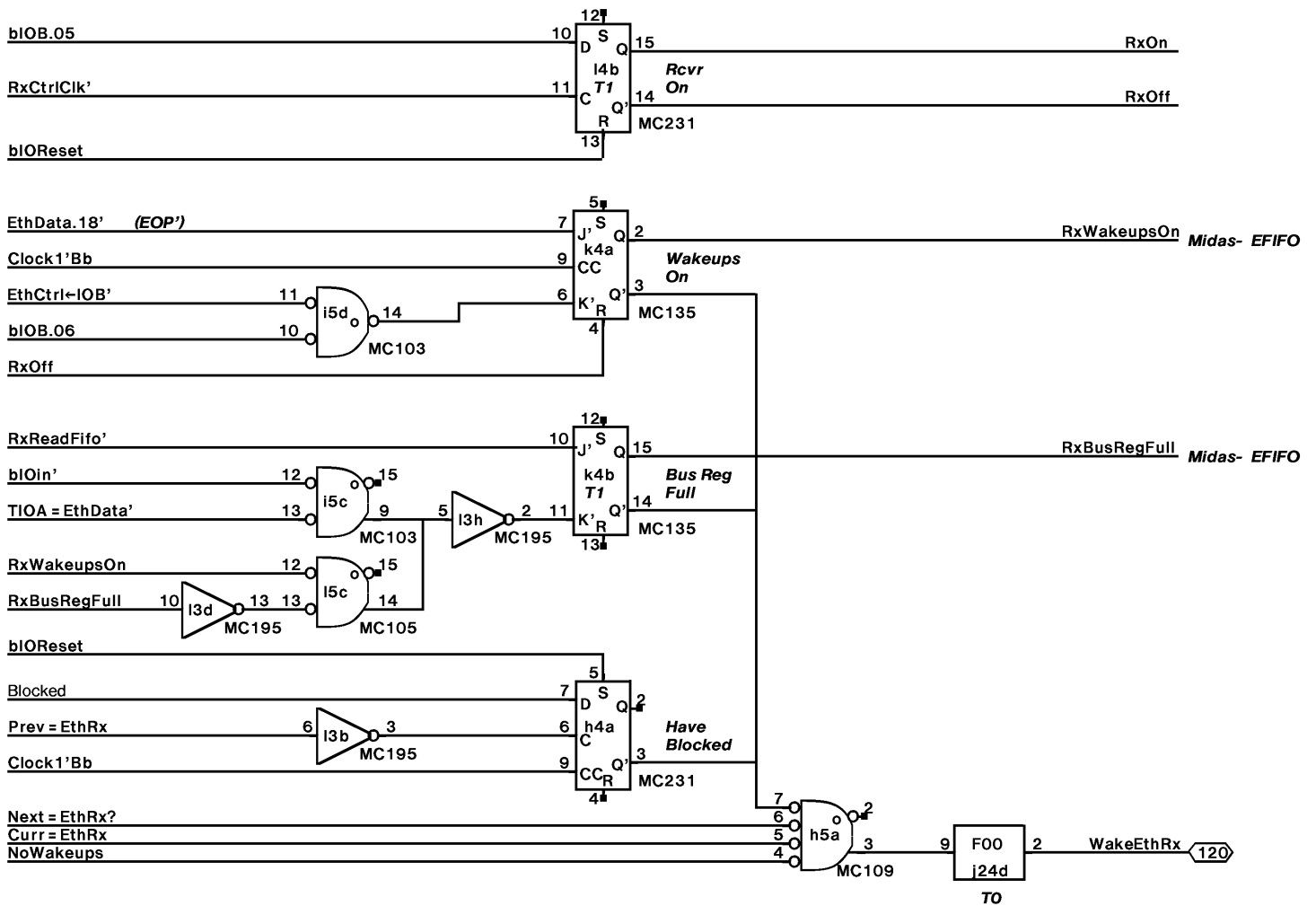




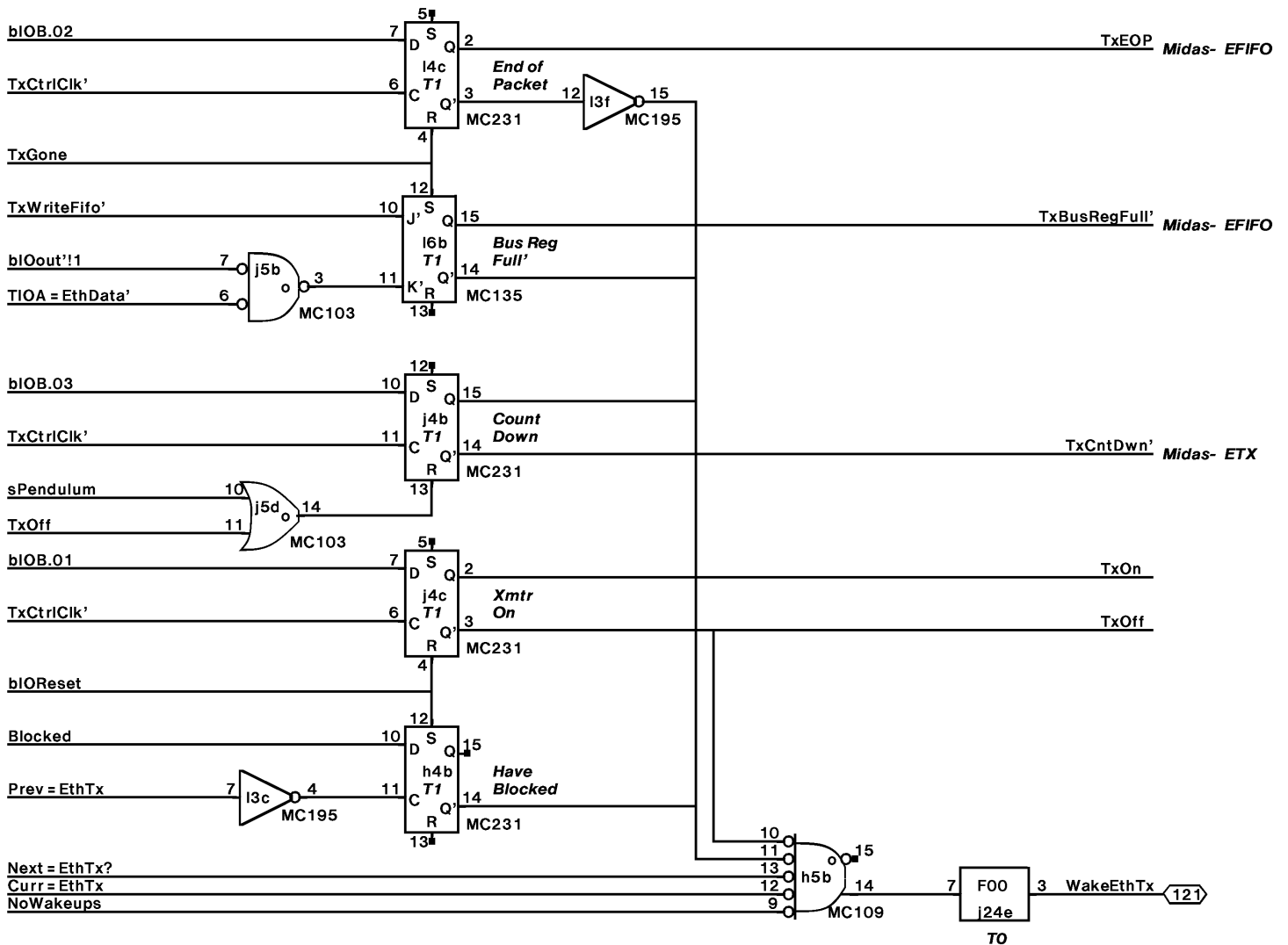
See Ether18.sil for timing diagrams



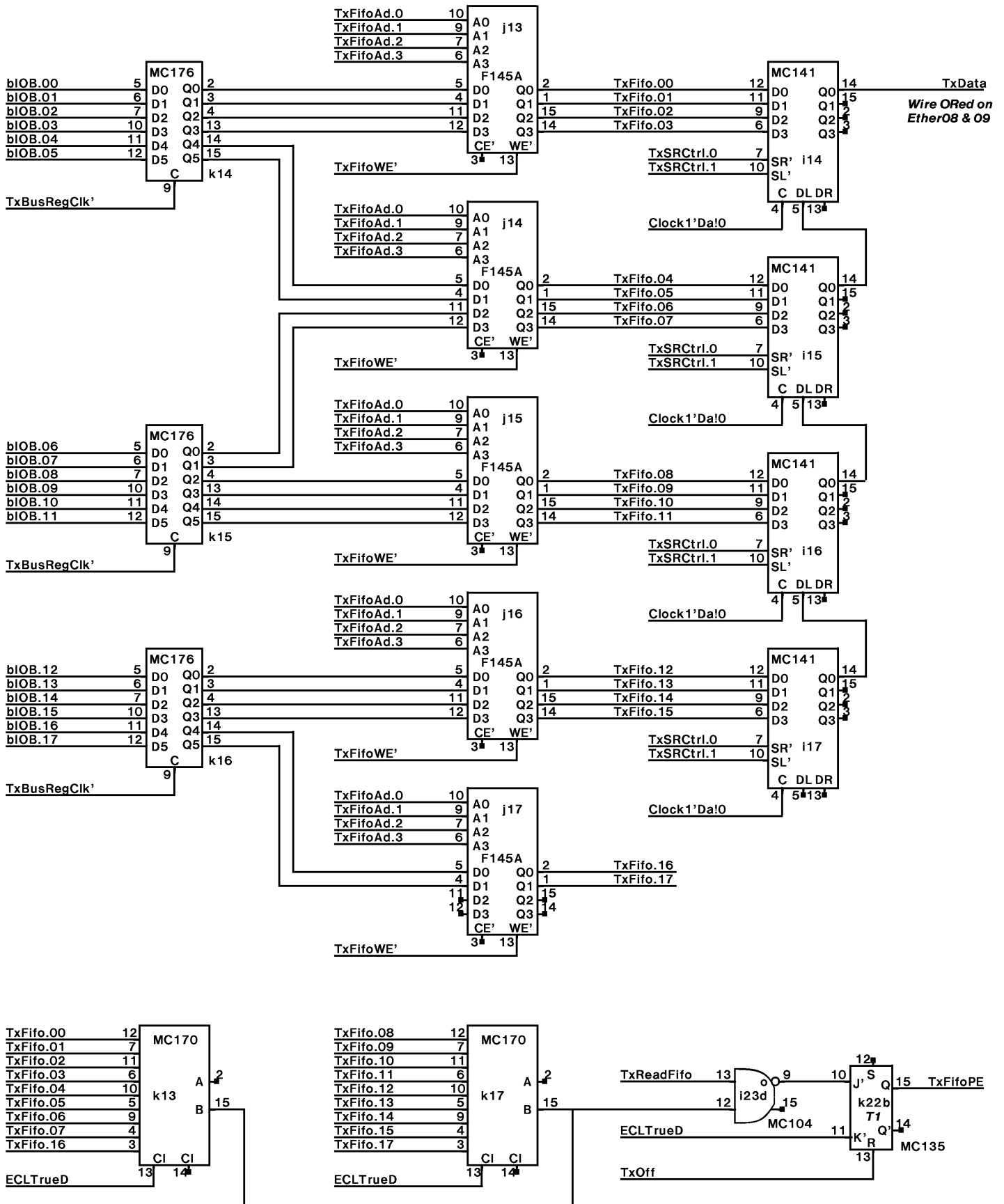


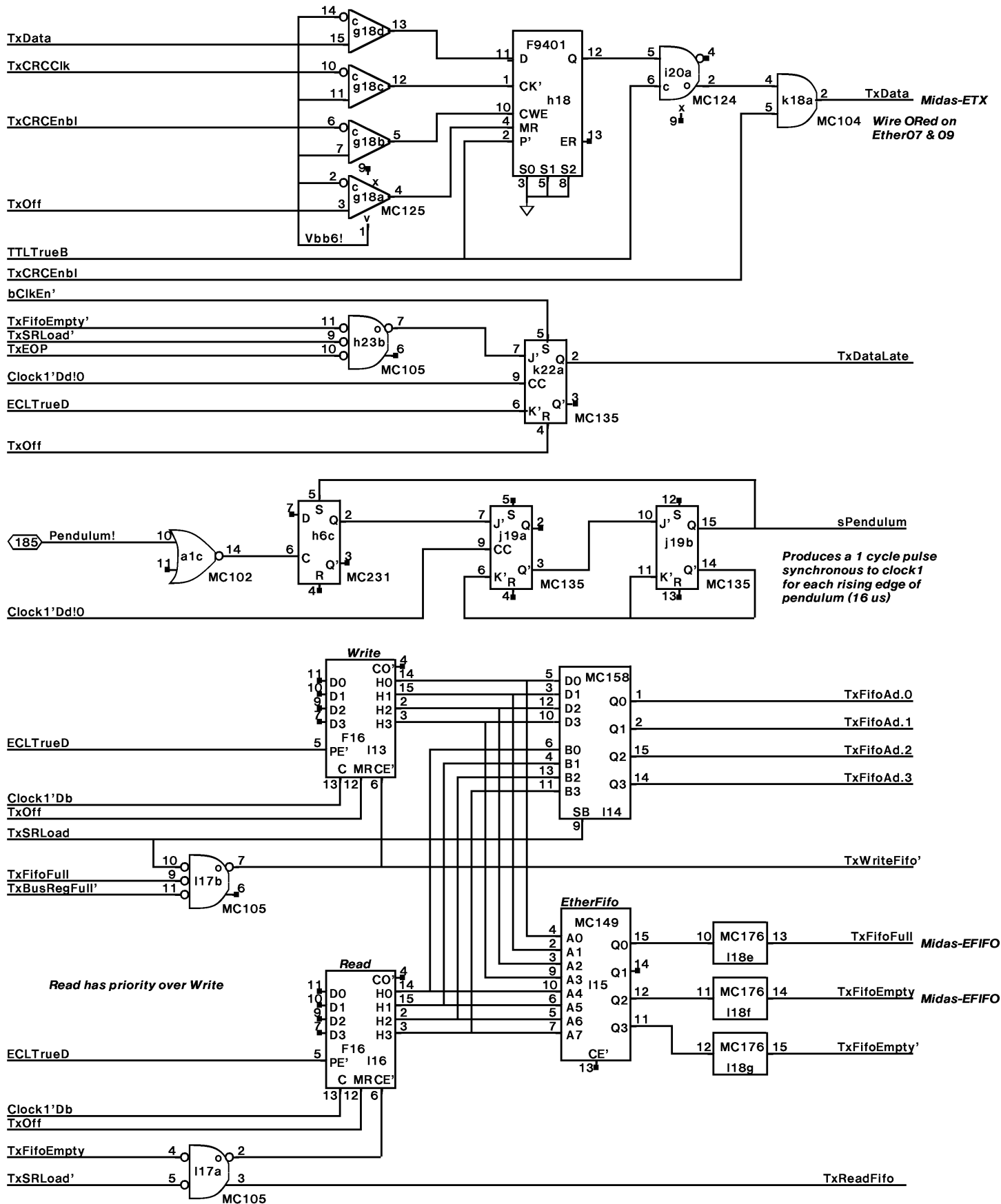


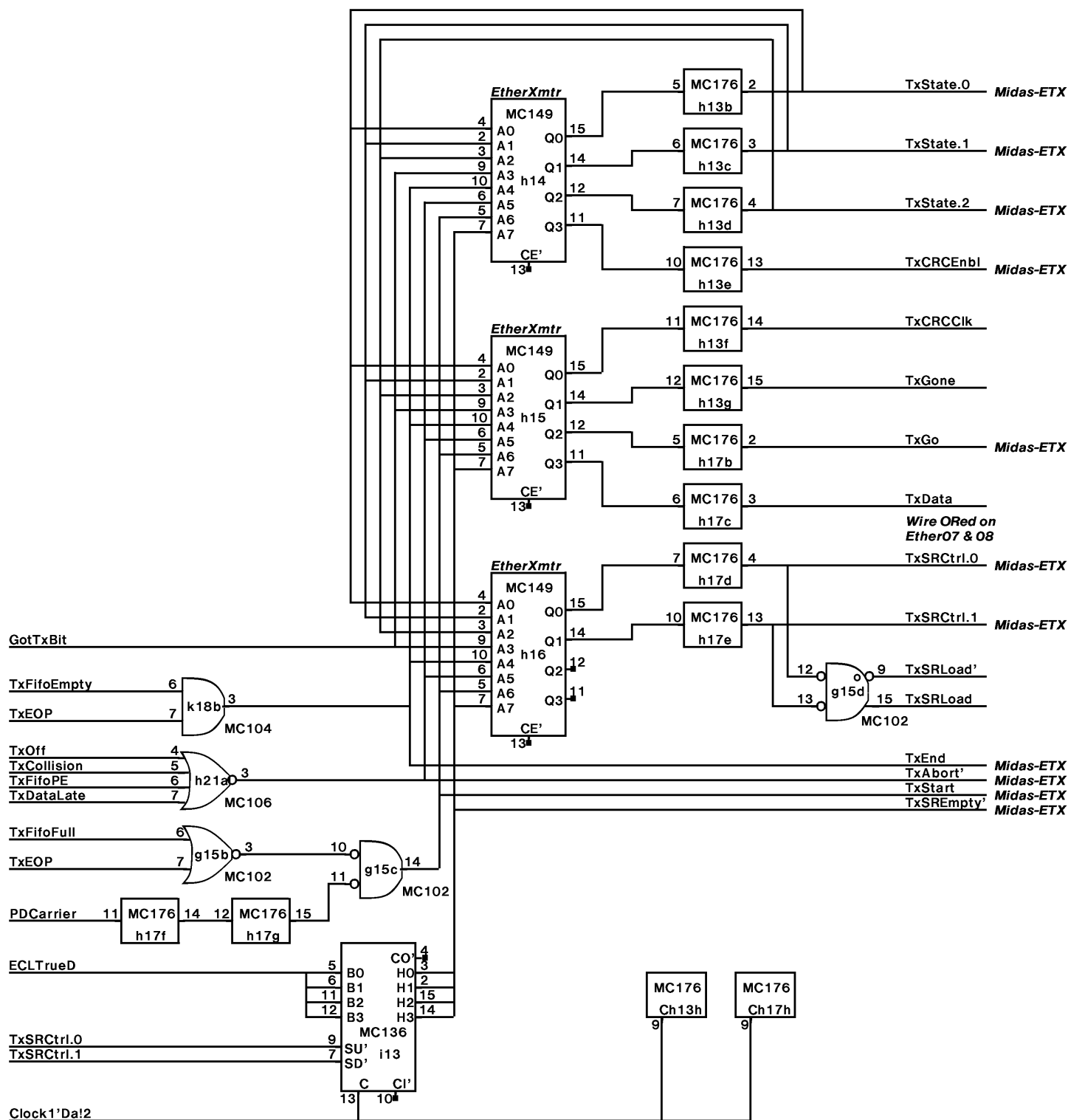
See Ether20.sil for wakeup timing diagram



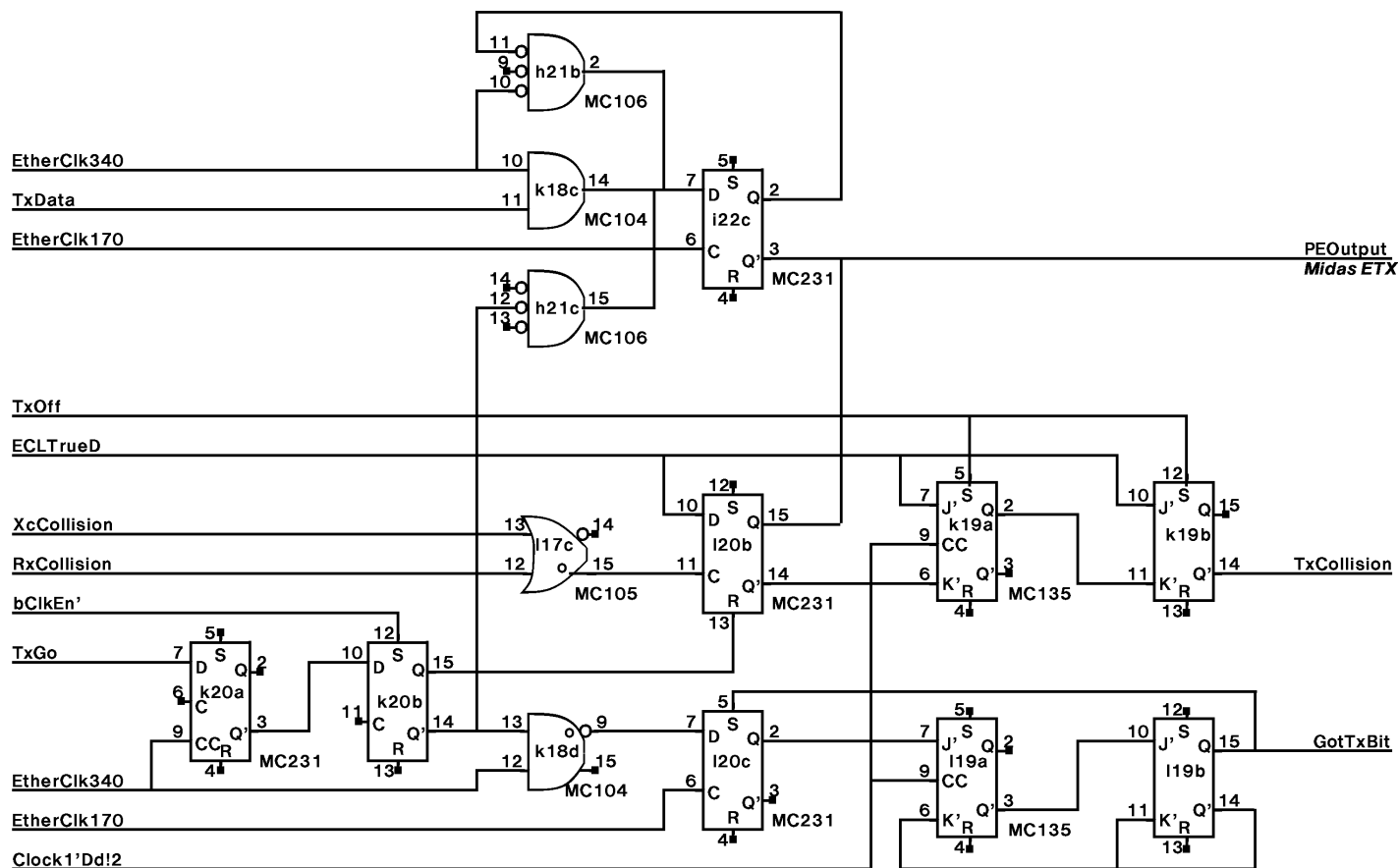
See Ether20.sil for wakeup timing diagram



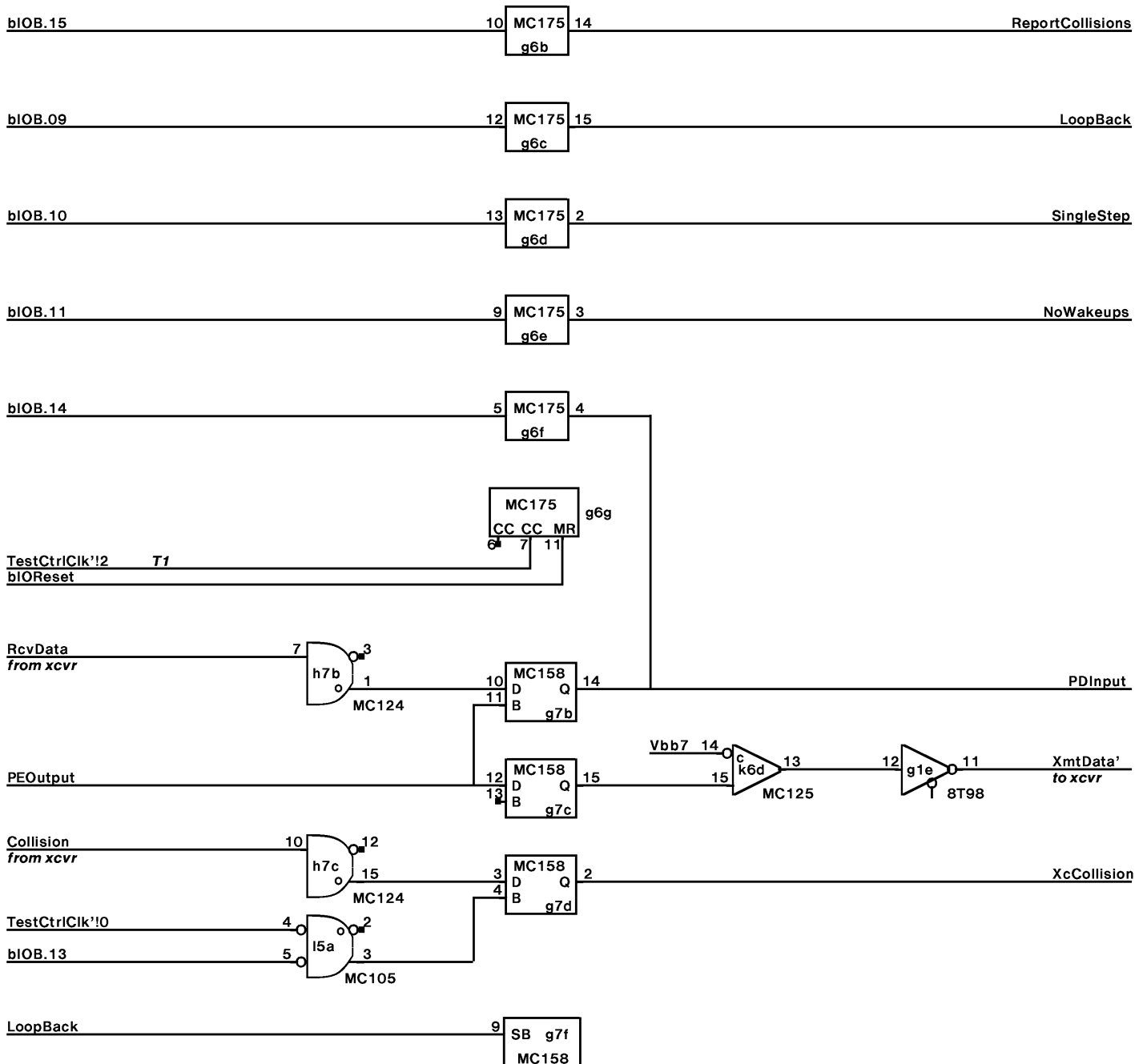


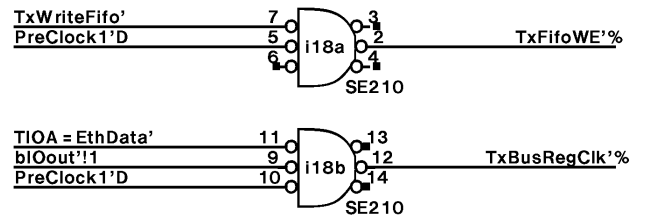
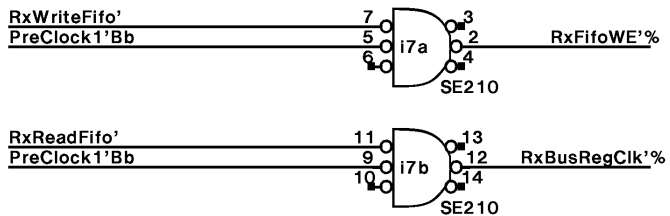
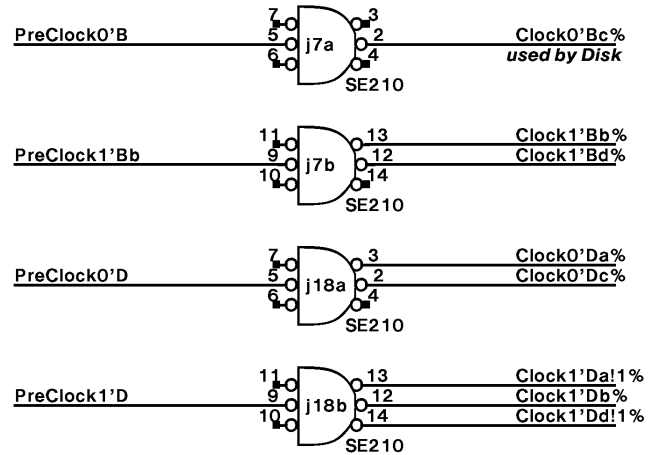
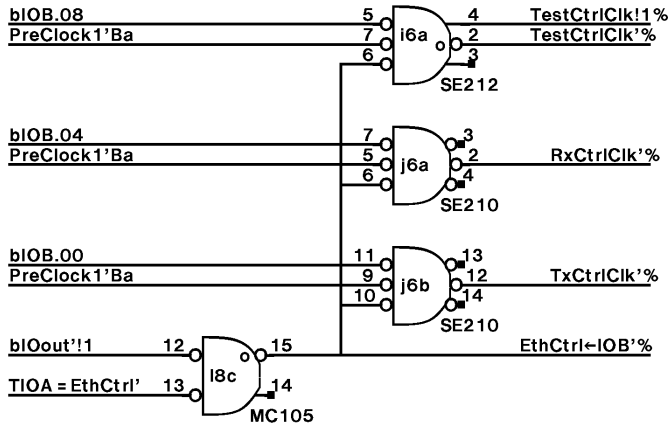


See Ether17.sil for timing diagrams



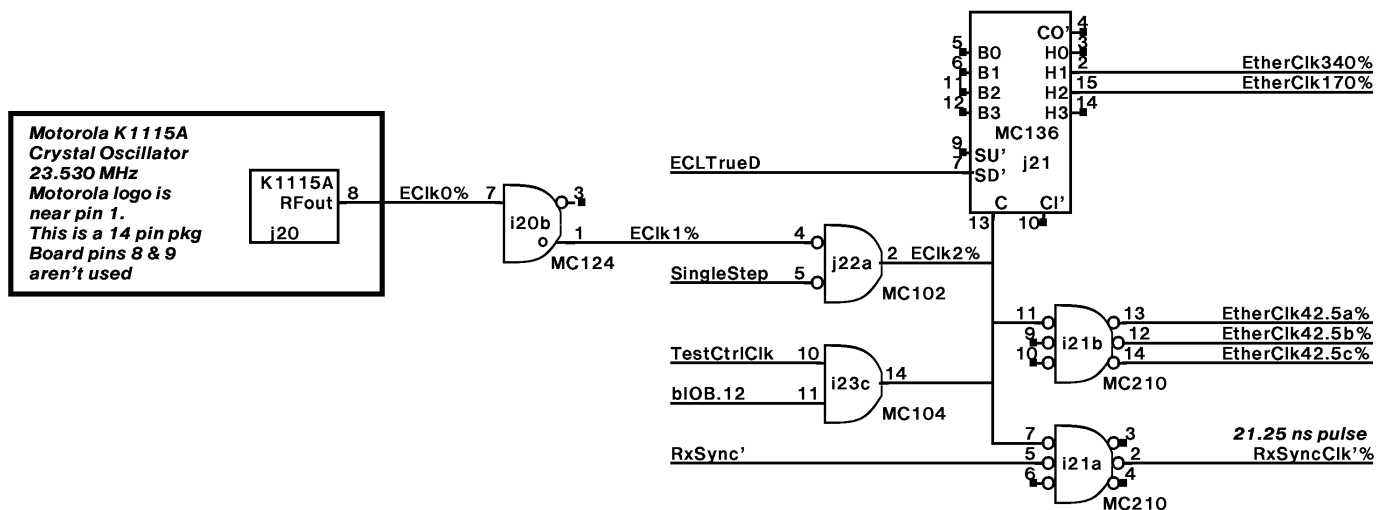
See Ether19.sil for timing diagrams

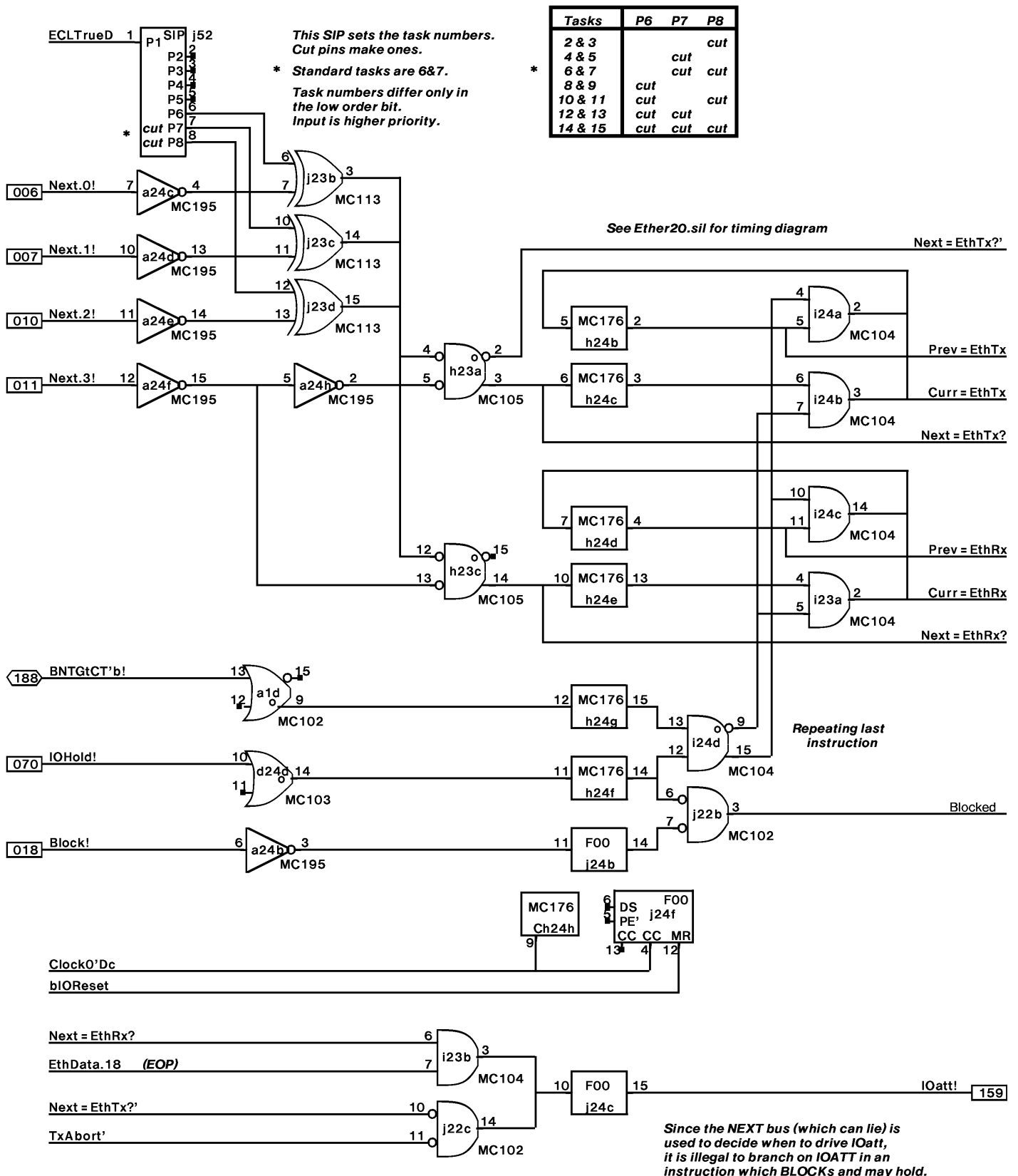


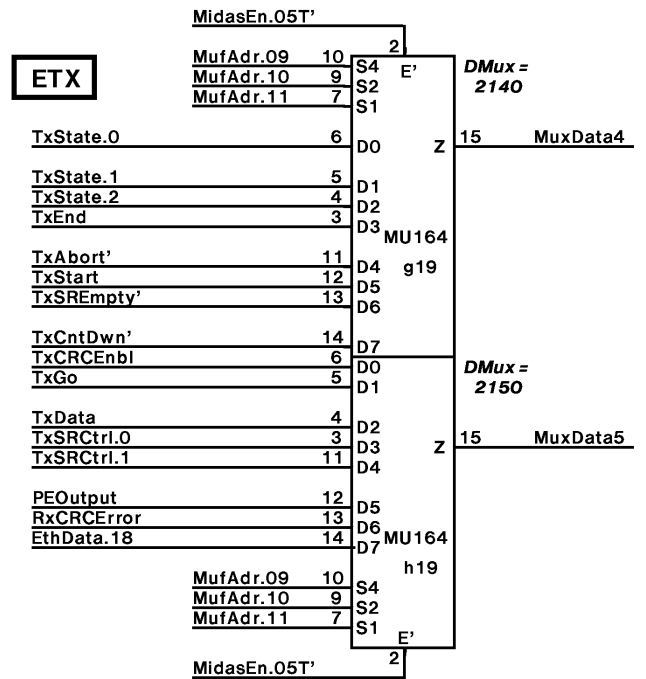
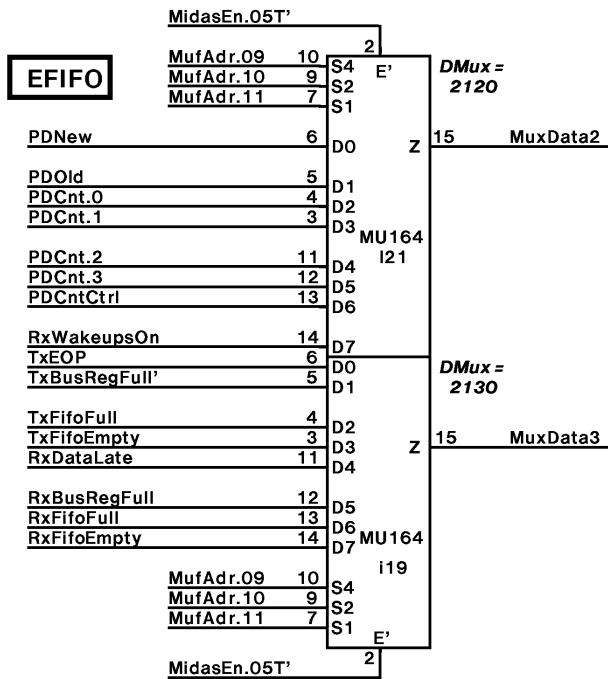


Dorado Synchronous Clocks

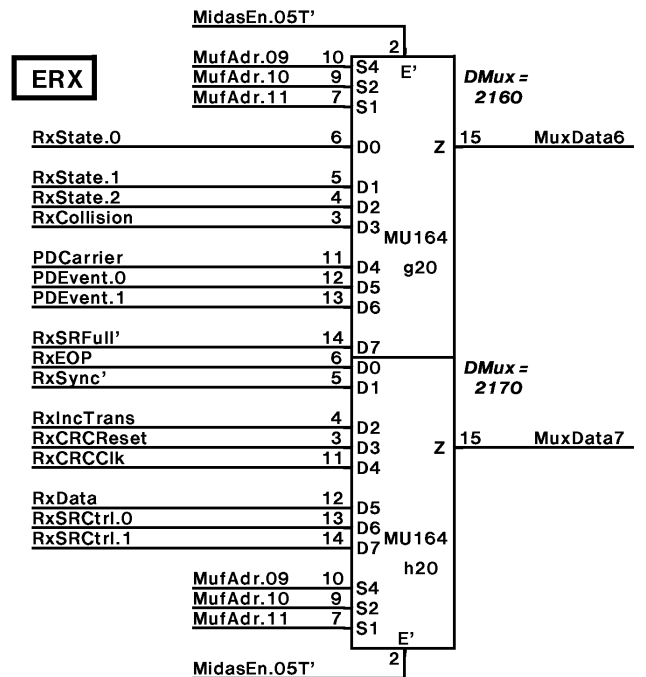
Free-running Ether Clocks





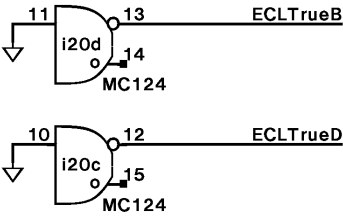
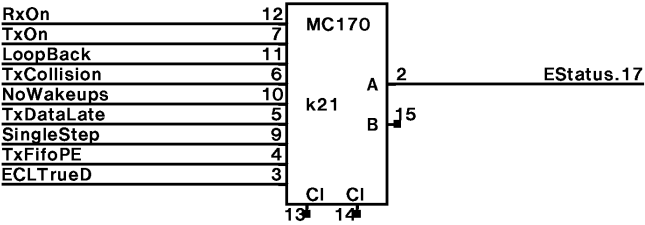
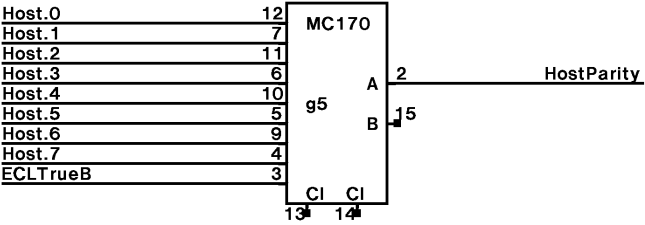
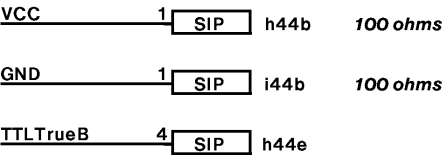
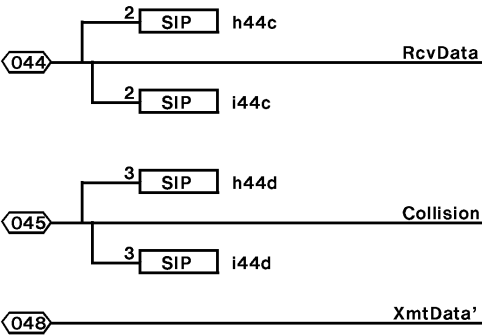


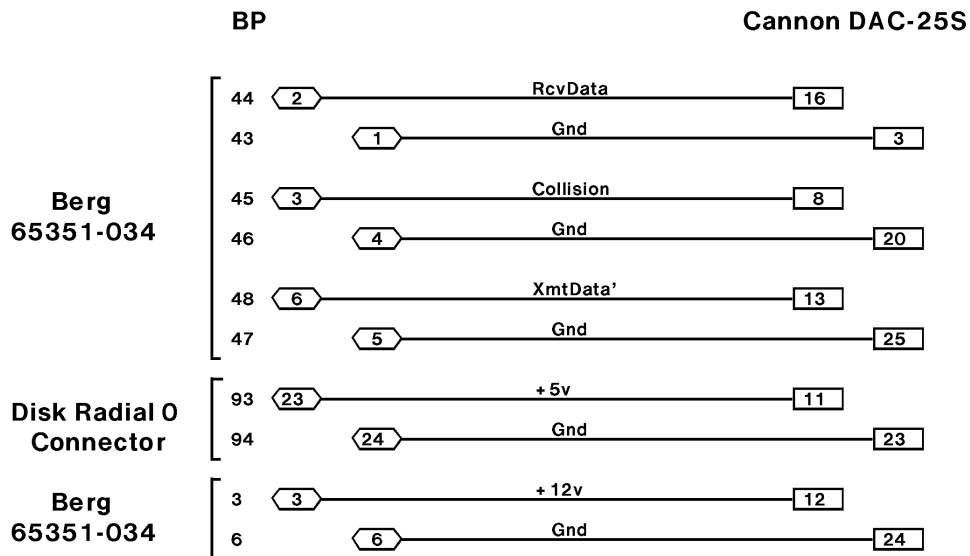
See DskEth01.sil for muffler control logic



To set a host address bit to 1
pull it up to gnd through 91 ohms.

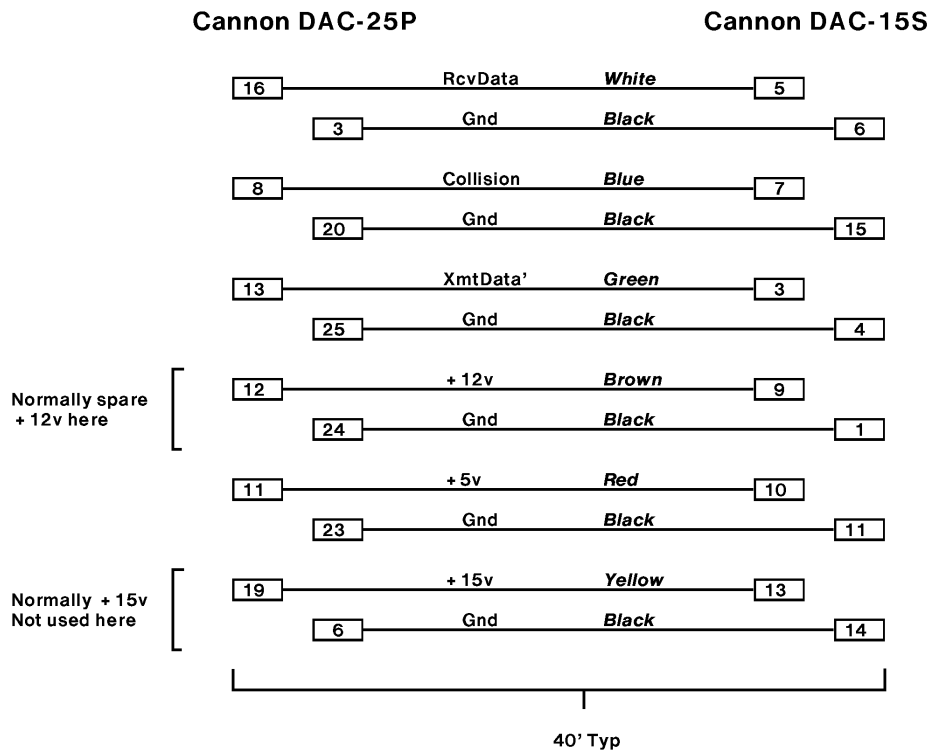
028	Host.0	200
029	Host.1	100
032	Host.2	40
033	Host.3	20
036	Host.4	10
037	Host.5	4
040	Host.6	2
041	Host.7	1
093	VCC	



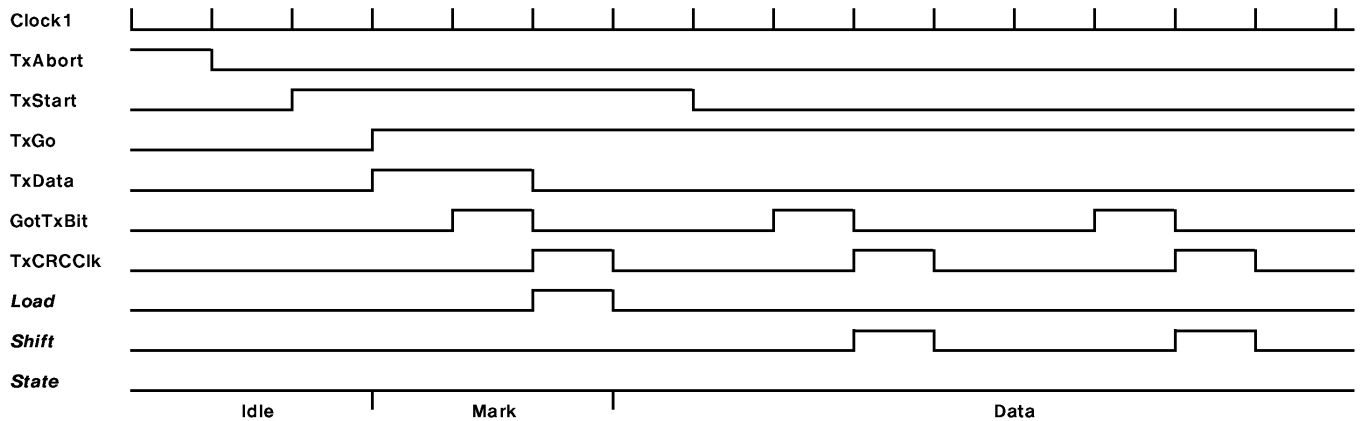


Internal Cable

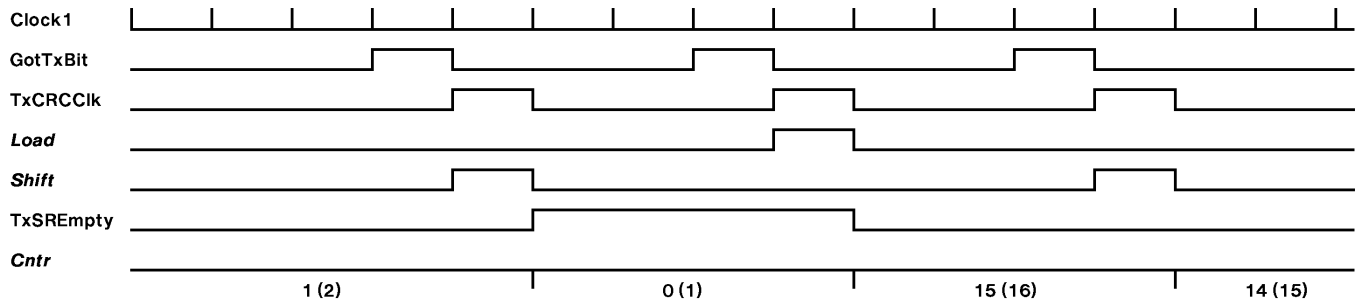
External Cable



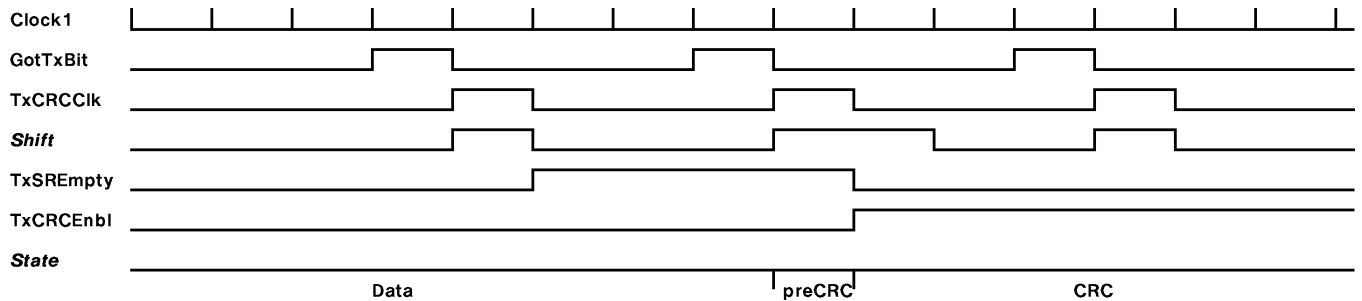
This is a standard Alto II Ethernet external cable part # 216411



Startup

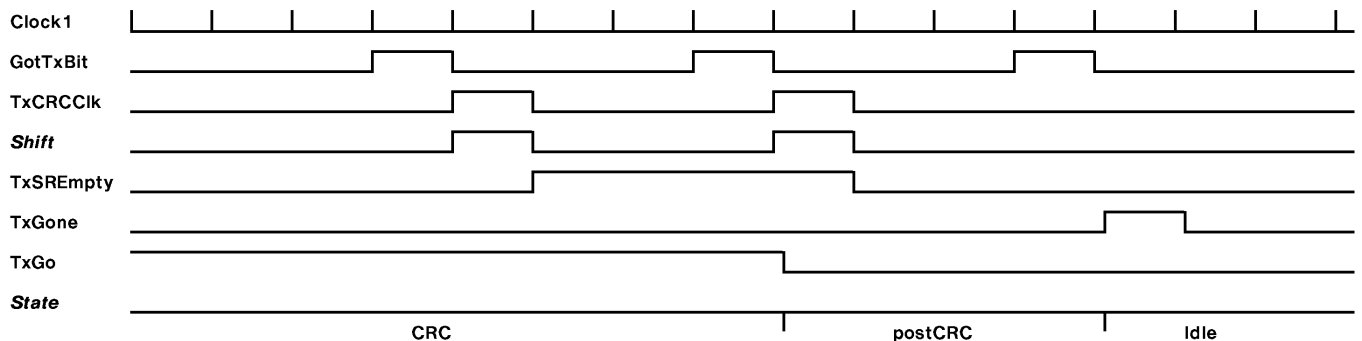


Loading Tx Shift Register

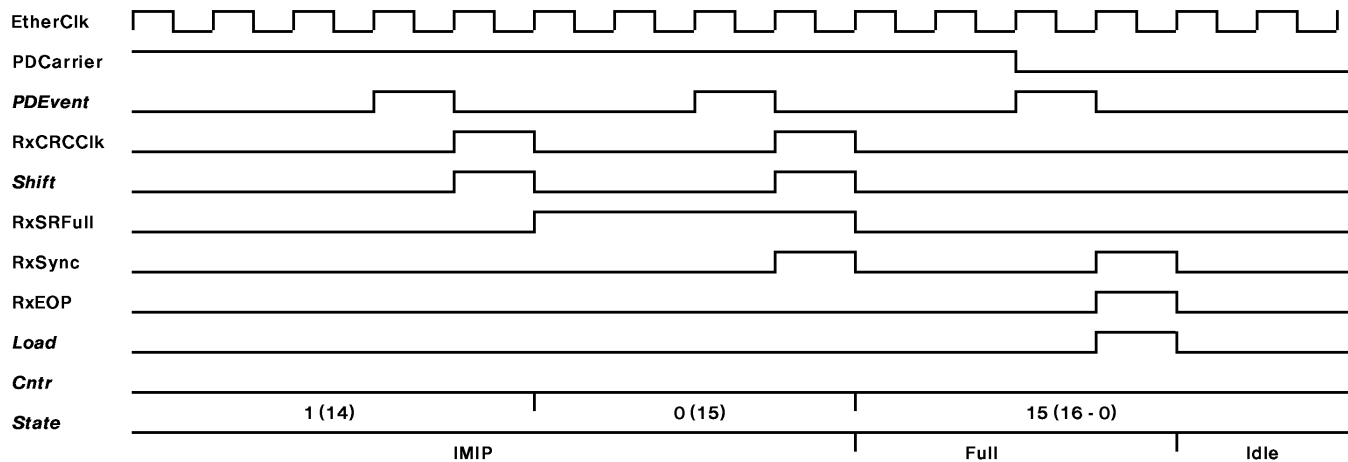
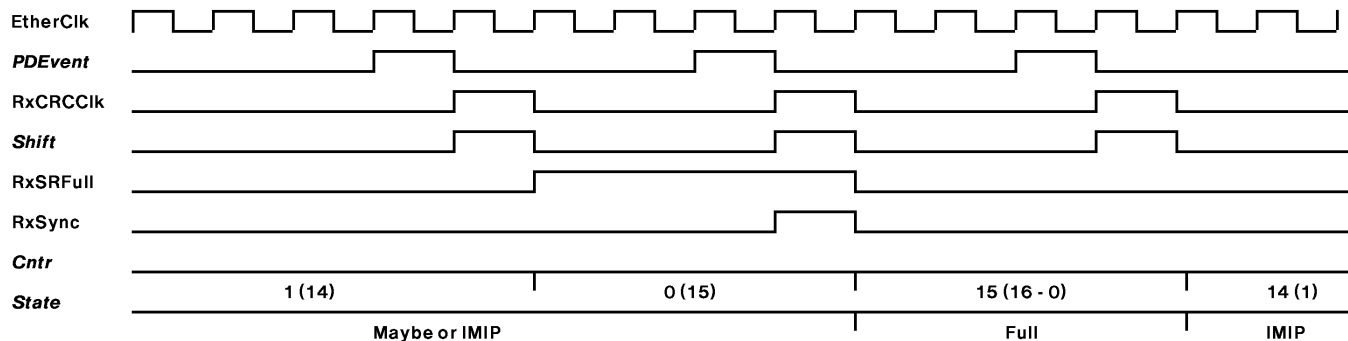
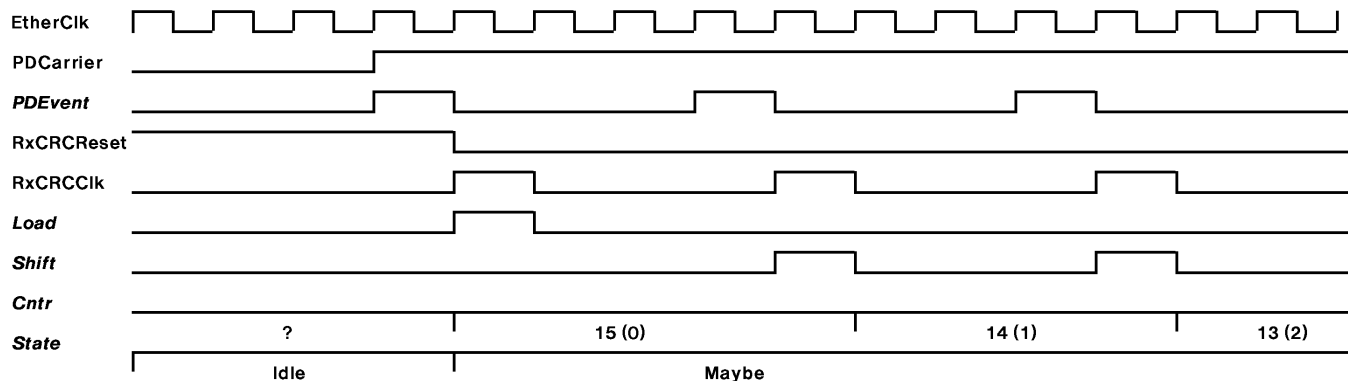


TxEnd has been high since TxSR was last loaded

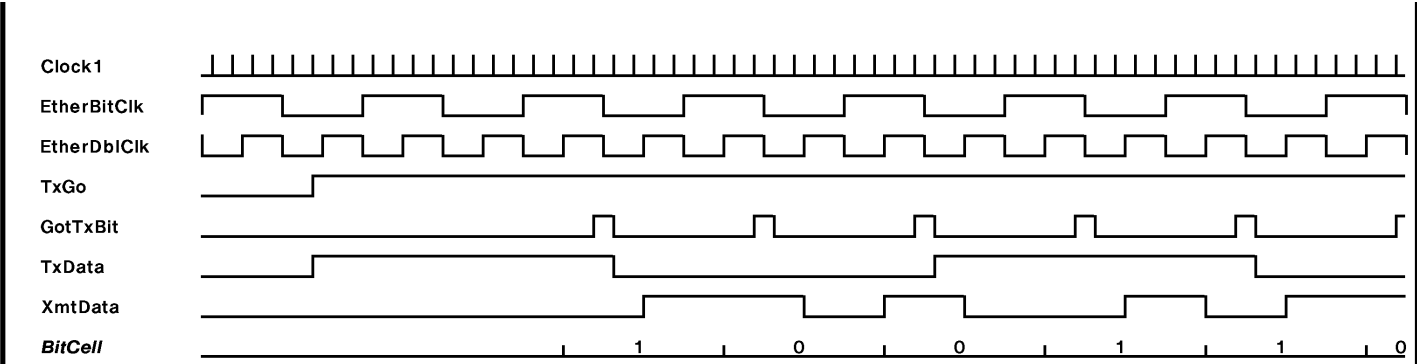
CRC



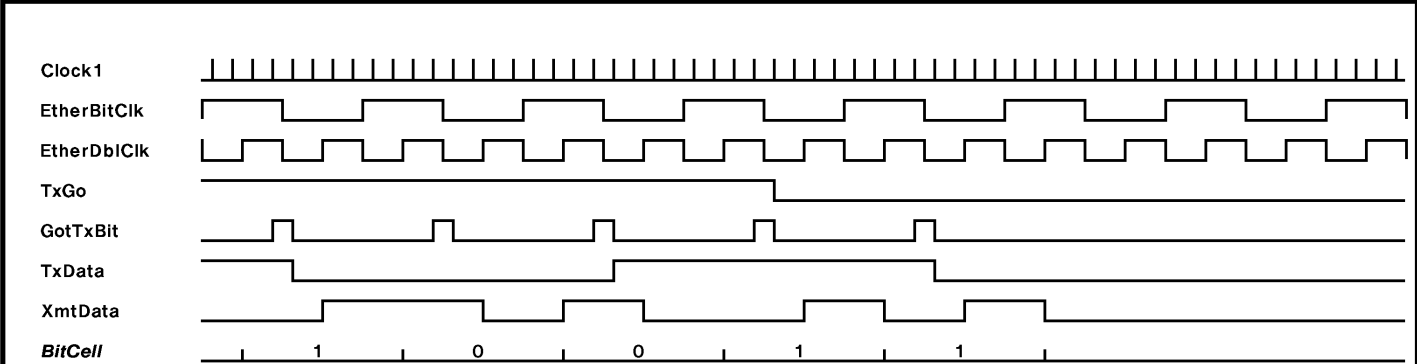
ShutDown



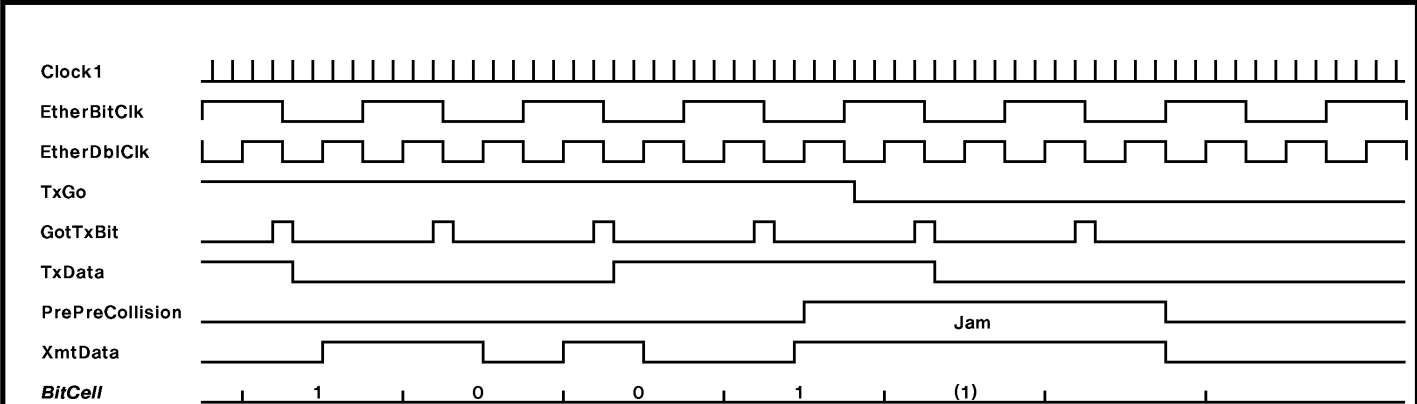
Cntr notation is <cntr value><# occupied postions>
 Phase decoder events are encoded in PDEvent.x
 Shift and Load are encoded in RxSRCtrl.x
 state is encoded in RxState.x
 shift also decrements Cntr



Startup

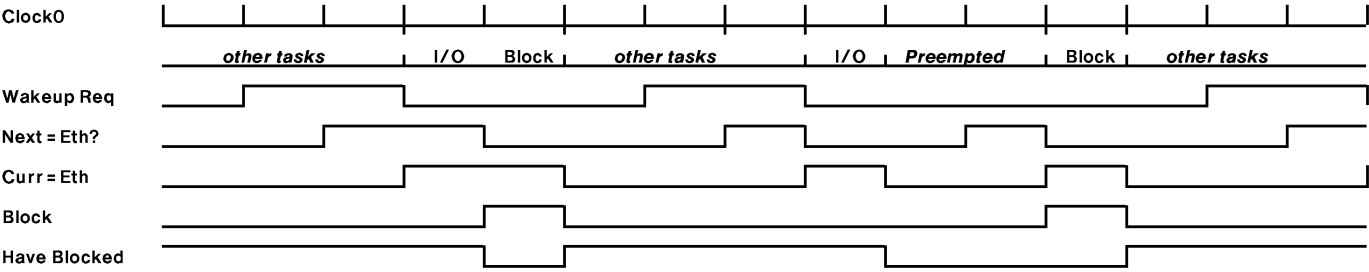


Shutdown



PrePreCollision is the output of the first stage of the Collision synchronizer

Collision



I/O means TIOA = EthData and FF = Input or Output

Wakeup Timing

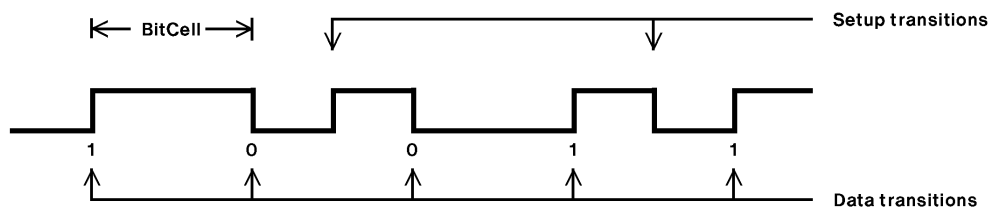
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Tx Cmd Enbl'	Tx On	Tx EOP	Tx Cnt Dwn	Rx Cmd Enbl'	Rx On	Rx BOP'		Test Cmd Enbl'	Loop Back	Single Step	No Wake ups	Test Clock	Test Coll'	Test Data	Report Colls

Output

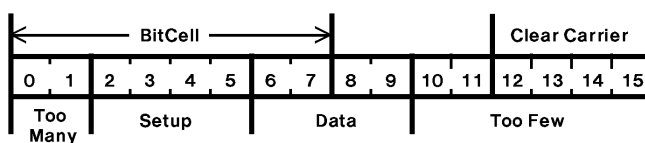
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Host Address								Rx On	Tx On	Loop Back	Tx Coll	No Wake ups	Tx Data Late	Single Step	Tx Fifo PE
028	029	032	033	036	037	040	041								

Input

The host address is set by jumpers on the right backplane.
 To set a bit to one, pull it up to ground through 91 ohms.



*A bitcell is nominally 340 ns.
A sample is nominally 42.5 ns.*



Inputs				Outputs			Comment
Carrier	Old	New	Cnt	Carrier	Event	CntCtrl	
Low	Low	Low	d/c	Low	Nothing	Count	Idle
Low	Low	High	d/c	High	One	Reset	Start of packet (start bit)
Low	High	Low	d/c	High	Collision	Reset	Impossible
Low	High	High	d/c	Low	Nothing	Count	Impossible
High	low	High	0-1	High	Collision	Count	Too many transitions
High	High	low	0-1	High	Collision	Count	Too many transitions
High	low	High	2-5	High	Nothing	Count	Setup transition (zero next)
High	High	low	2-5	High	Nothing	Count	Setup transition (one next)
High	low	High	6-9	High	One	Reset	Data transition
High	High	low	6-9	High	Zero	Reset	Data transition
High	low	High	10-15	High	Collision	Reset	Too few transitions
High	High	low	10-15	High	Collision	Reset	Too few transitions
High	Low	Low	0-11	High	Nothing	Count	Active
High	High	High	0-11	High	Nothing	Count	Active
High	Low	Low	12-15	Low	Collision	Reset	End of packet
High	High	High	12-15	High	Collision	Reset	Jam

*"Impossible" conditions can happen right after power up.
d/c means "don't care".*